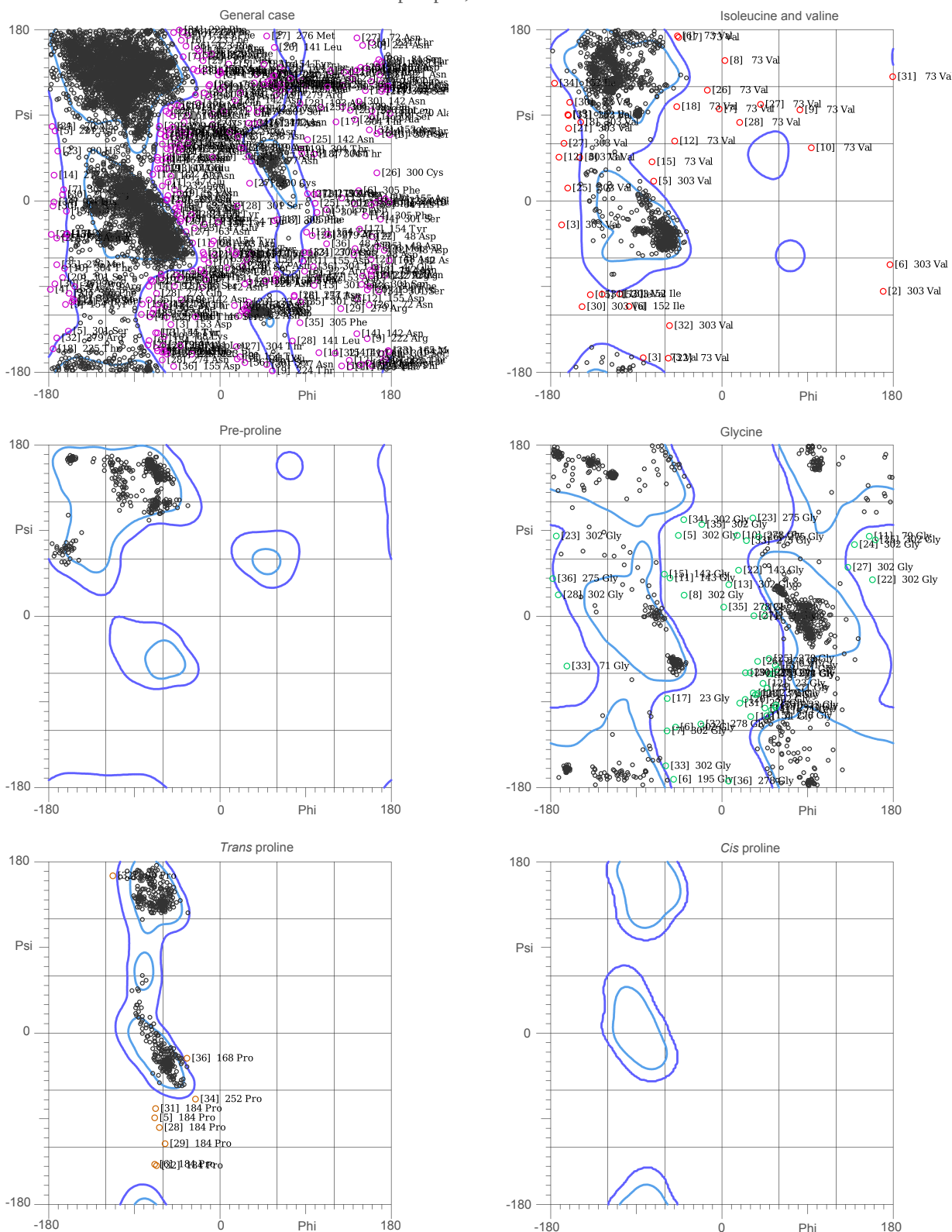


MolProbity Ramachandran analysis

7mhp.H.pdb, all models



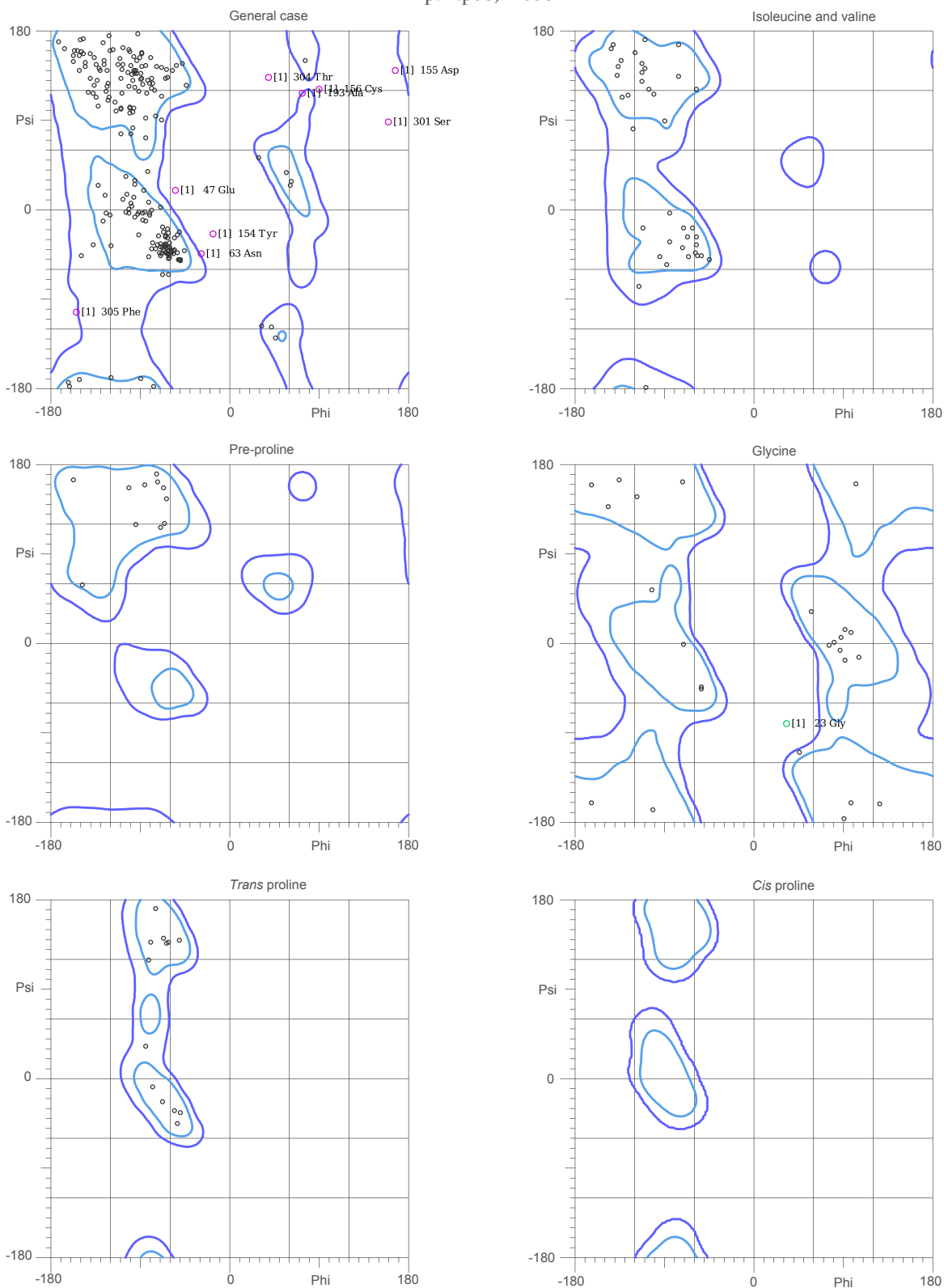
87.9% (9624/10944) of all residues were in favored (98%) regions.
 95.9% (10496/10944) of all residues were in allowed (>99.8%) regions.
 This list is truncated; use the MolProbity multi-chart.html for complete list.
 There were 448 outliers (phi, psi):

[1] 23 Gly (33.4, -81.7)	[1] 304 Thr (39.7, 134.3)	[2] 193 Ala (67.1, 138.5)
[1] 47 Glu (-55.9, 20.1)	[1] 305 Phe (-155.4, -103.9)	[2] 273 Gln (-73.2, -83.4)
[1] 63 Asn (-29.1, -44.1)	[2] 47 Glu (-48.5, 12.2)	[2] 279 Arg (-15.5, 154.2)
[1] 154 Tyr (-17.9, -24.7)	[2] 71 Gly (45.0, 9.8)	[2] 303 Val (170.3, -95.8)
[1] 155 Asp (167.2, 141.2)	[2] 72 Asn (174.4, -78.2)	[2] 304 Thr (-56.1, -158.6)
[1] 156 Cys (90.3, 122.7)	[2] 153 Asp (-2.6, 133.6)	[3] 33 Asp (28.6, -110.0)
[1] 193 Ala (73.9, 118.4)	[2] 154 Tyr (31.8, -85.9)	[3] 63 Asn (-17.9, -61.6)
[1] 301 Ser (160.2, 89.6)	[2] 155 Asp (-178.3, -35.5)	[3] 72 Asn (160.5, -74.7)
		[3] 73 Val (-83.7, -165.2)
		[3] 153 Asp (-52.6, -130.7)
		[3] 154 Tyr (21.3, -164.5)
		[3] 225 Thr (174.6, -167.2)

[3] 279 Arg (-61.5, -159.5)	[4] 301 Ser (169.6, -19.3)	[5] 302 Gly (-46.9, 85.3)
[3] 301 Ser (179.4, 70.8)	[5] 63 Asn (-52.4, 45.8)	[5] 303 Val (-72.3, 21.2)
[3] 303 Val (-169.8, -25.5)	[5] 73 Val (-150.2, 46.8)	[6] 46 Ser (-66.9, 43.2)
[4] 33 Asp (25.8, -114.9)	[5] 74 Gln (-8.1, 92.2)	[6] 63 Asn (-43.3, -6.2)
[4] 153 Asp (-179.3, -93.7)	[5] 153 Asp (-18.7, -54.1)	[6] 72 Asn (-46.5, -79.0)
[4] 154 Tyr (-6.1, -57.4)	[5] 154 Tyr (-7.6, -42.1)	[6] 73 Val (-47.5, 174.4)
[4] 185 Phe (-24.8, 153.7)	[5] 155 Asp (112.8, -9.4)	[6] 142 Asn (-19.7, -92.0)
[4] 223 Phe (-74.1, -122.3)	[5] 184 Pro (-69.7, -90.0)	[6] 72 Ile (-98.3, -111.2)
[4] 224 Thr (160.9, 167.0)	[5] 185 Phe (63.1, 124.0)	[6] 153 Asp (103.4, -81.7)
[4] 232 Leu (-61.1, 16.5)	[5] 221 Asn (-174.7, 74.7)	[6] 154 Tyr (-75.7, -147.6)
[4] 277 Asn (-42.6, 66.8)	[5] 222 Arg (-1.1, 74.8)	[6] 184 Pro (-69.3, -138.8)
[4] 278 Gly (45.3, 1.0)	[5] 301 Ser (-160.1, -137.5)	[6] 185 Phe (85.6, 127.3)

MolProbity Ramachandran analysis

7mhp.H.pdb, model 1



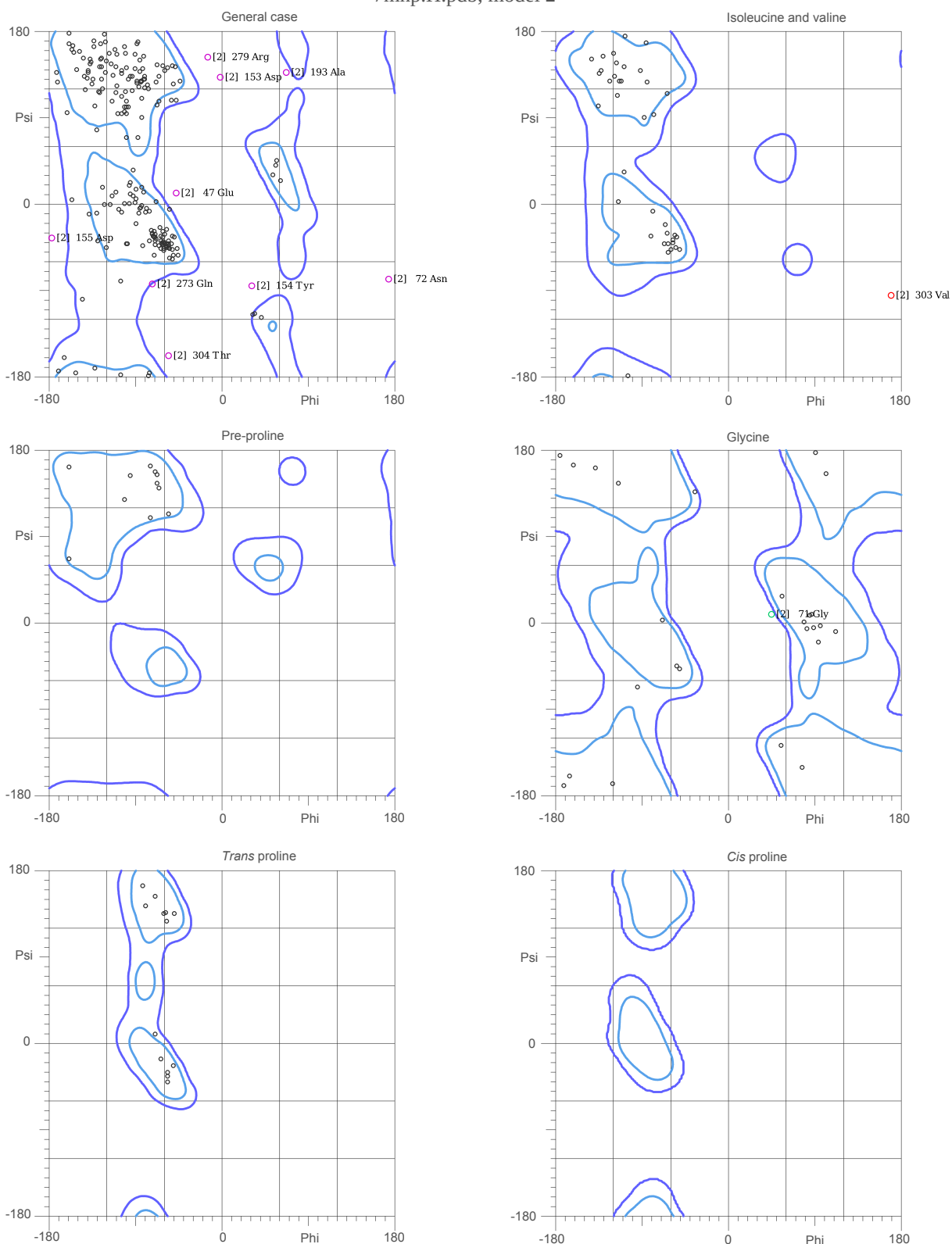
90.1% (274/304) of all residues were in favored (98%) regions.
96.7% (294/304) of all residues were in allowed (>99.8%) regions.

There were 10 outliers (phi, psi):

[1] 23 Gly (33.4, -81.7)	[1] 304 Thr (39.7, 134.3)
[1] 47 Glu (-55.9, 20.1)	[1] 305 Phe (-155.4, -103.9)
[1] 63 Asn (-29.1, -44.1)	
[1] 154 Tyr (-17.9, -24.7)	
[1] 155 Asp (167.2, 141.2)	
[1] 156 Cys (90.3, 122.7)	
[1] 193 Ala (73.9, 118.4)	
[1] 301 Ser (160.2, 89.6)	

MolProbity Ramachandran analysis

7mhp.H.pdb, model 2



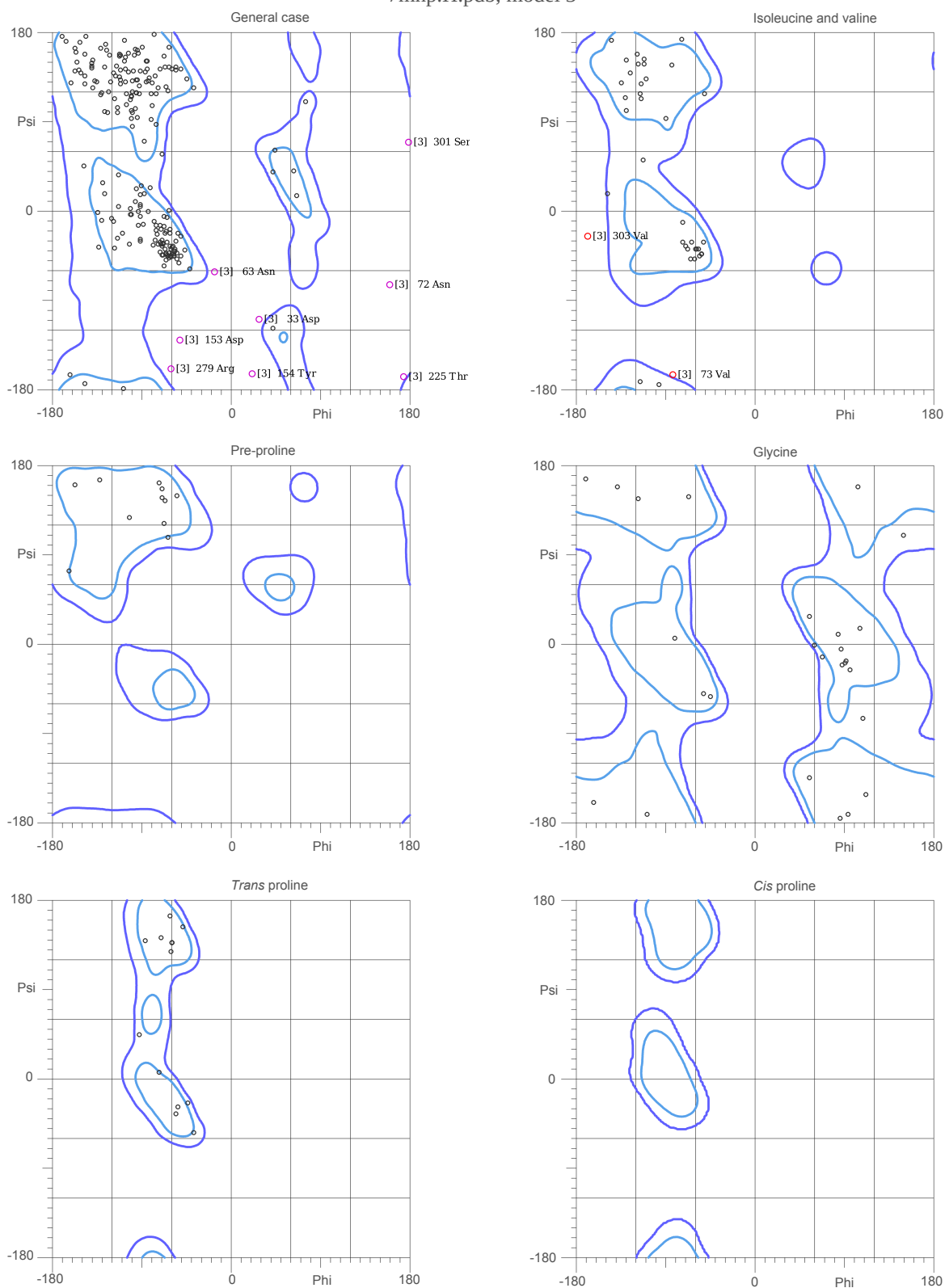
87.2% (265/304) of all residues were in favored (98%) regions.
96.4% (293/304) of all residues were in allowed (>99.8%) regions.

There were 11 outliers (phi, psi):

[2] 47 Glu (-48.5, 12.2)	[2] 279 Arg (-15.5, 154.2)
[2] 71 Gly (45.0, 9.8)	[2] 303 Val (170.3, -95.8)
[2] 72 Asn (174.4, -78.2)	[2] 304 Thr (-56.1, -158.6)
[2] 153 Asp (-2.6, 133.6)	
[2] 154 Tyr (31.8, -85.9)	
[2] 155 Asp (-178.3, -35.5)	
[2] 193 Ala (67.1, 138.5)	
[2] 273 Gln (-73.2, -83.4)	

MolProbity Ramachandran analysis

7mhp.H.pdb, model 3



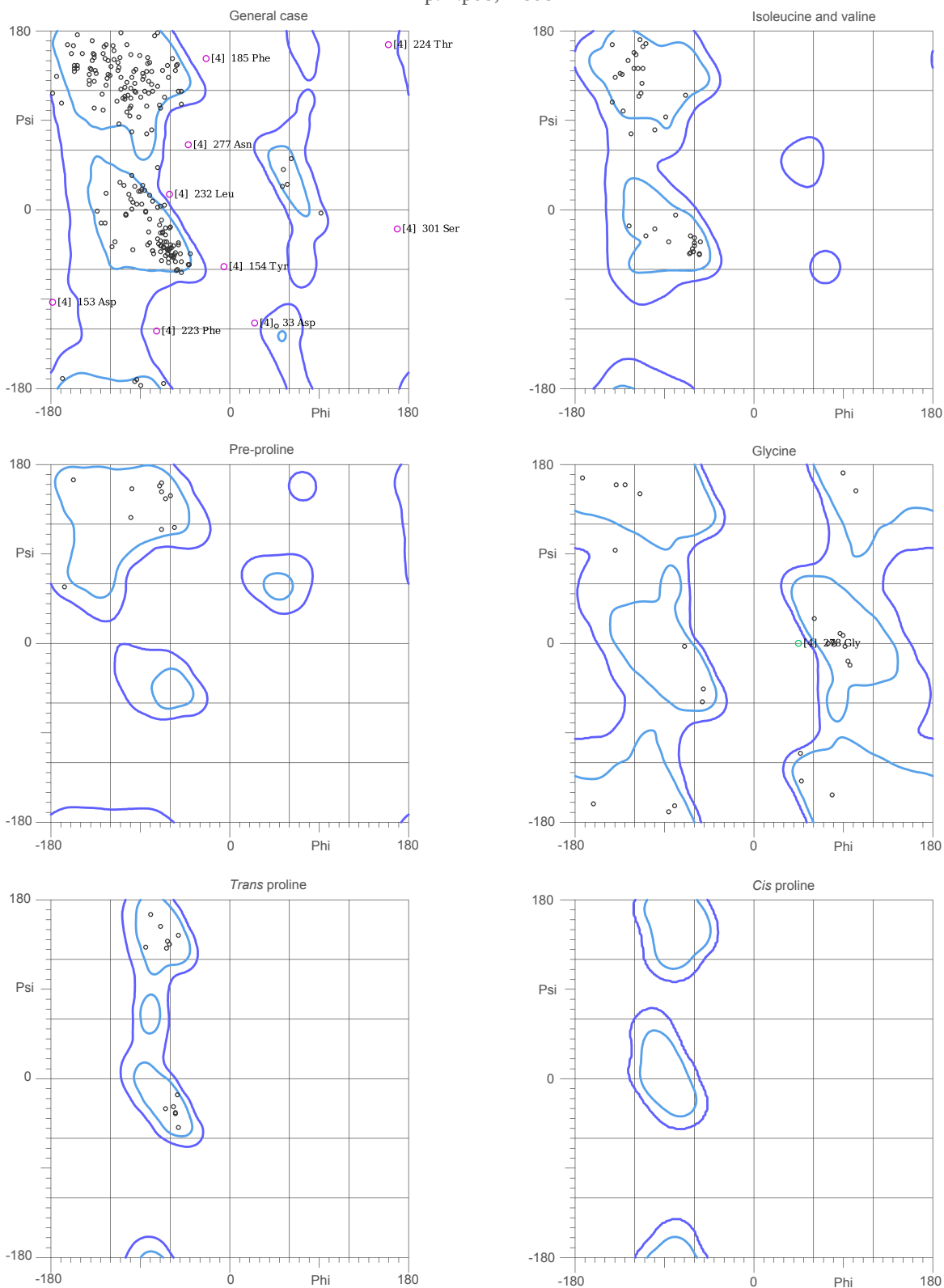
88.2% (268/304) of all residues were in favored (98%) regions.
96.7% (294/304) of all residues were in allowed (>99.8%) regions.

There were 10 outliers (phi, psi):

[3] 33 Asp (28.6, -110.0)	[3] 301 Ser (179.4, 70.8)
[3] 63 Asn (-17.9, -61.6)	[3] 303 Val (-169.8, -25.5)
[3] 72 Asn (160.5, -74.7)	
[3] 73 Val (-83.7, -165.2)	
[3] 153 Asp (-52.6, -130.7)	
[3] 154 Tyr (21.3, -164.5)	
[3] 225 Thr (174.6, -167.2)	
[3] 279 Arg (-61.5, -159.5)	

MolProbity Ramachandran analysis

7mhp.H.pdb, model 4



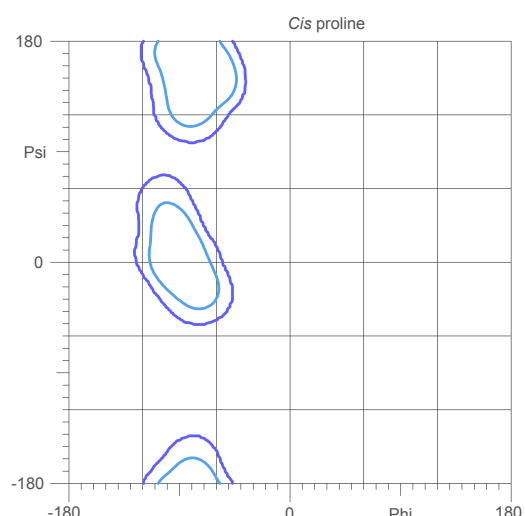
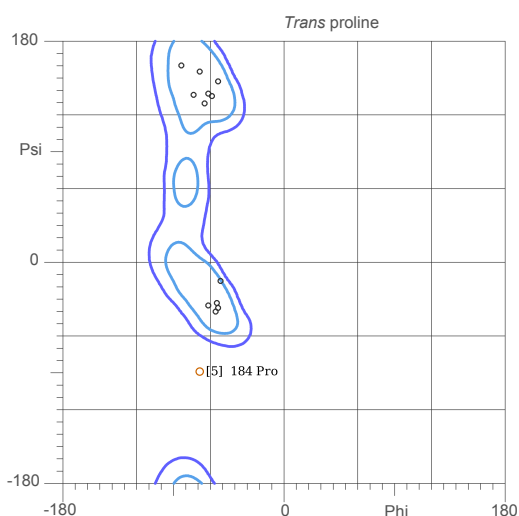
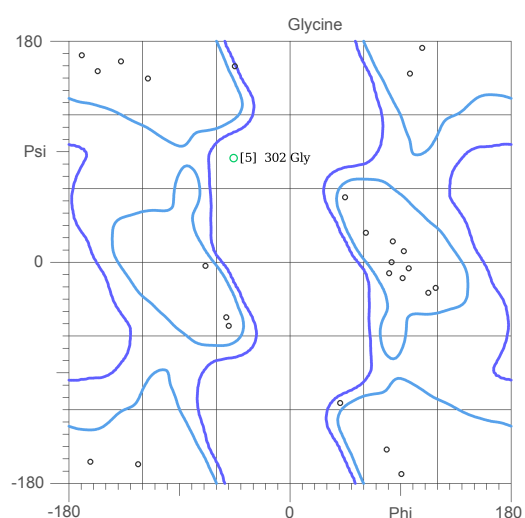
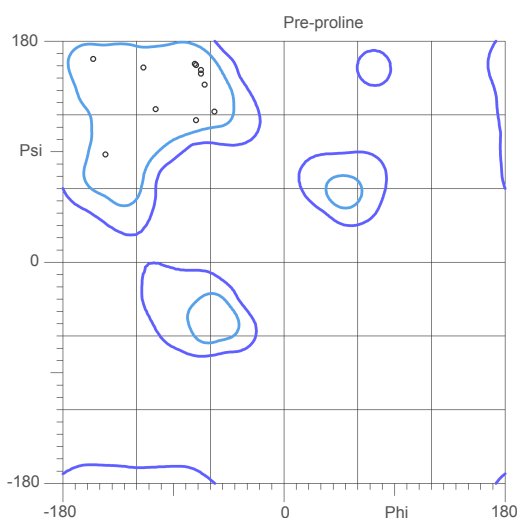
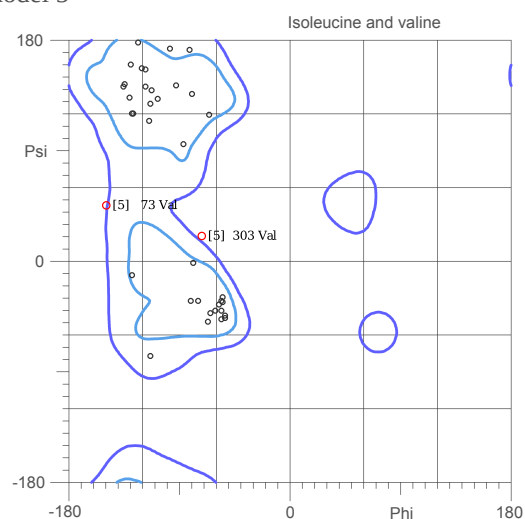
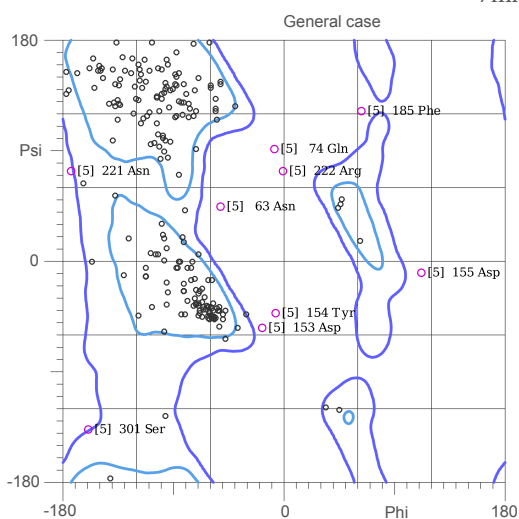
90.5% (275/304) of all residues were in favored (98%) regions.
96.7% (294/304) of all residues were in allowed (>99.8%) regions.

There were 10 outliers (phi, psi):

[4] 33 Asp (25.8, -114.9)	[4] 278 Gly (45.3, 1.0)
[4] 153 Asp (-179.3, -93.7)	[4] 301 Ser (169.6, -19.3)
[4] 154 Tyr (-6.1, -57.4)	
[4] 185 Phe (-24.8, 153.7)	
[4] 223 Phe (-74.1, -122.3)	
[4] 224 Thr (160.9, 167.0)	
[4] 232 Leu (-61.1, 16.5)	
[4] 277 Asn (-42.6, 66.8)	

MolProbity Ramachandran analysis

7mhp.H.pdb, model 5



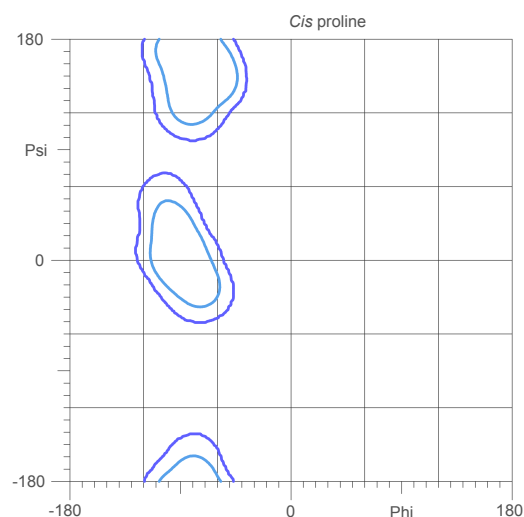
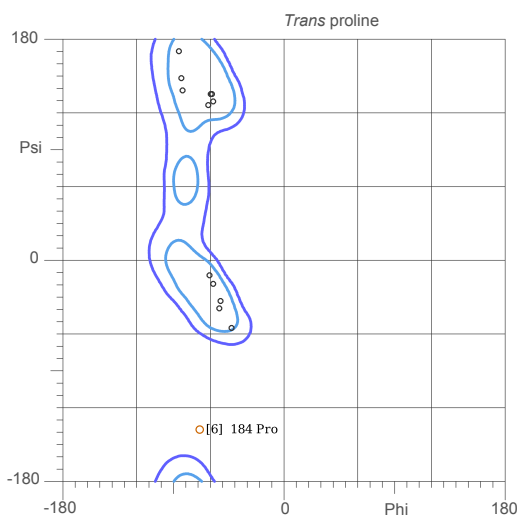
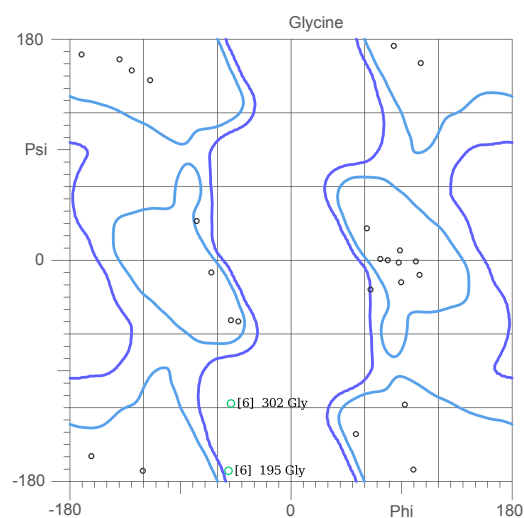
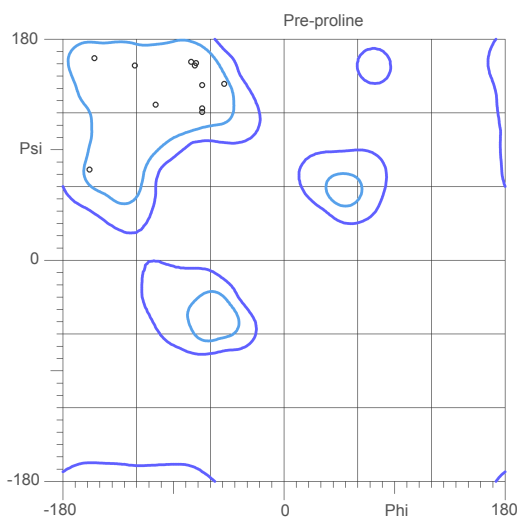
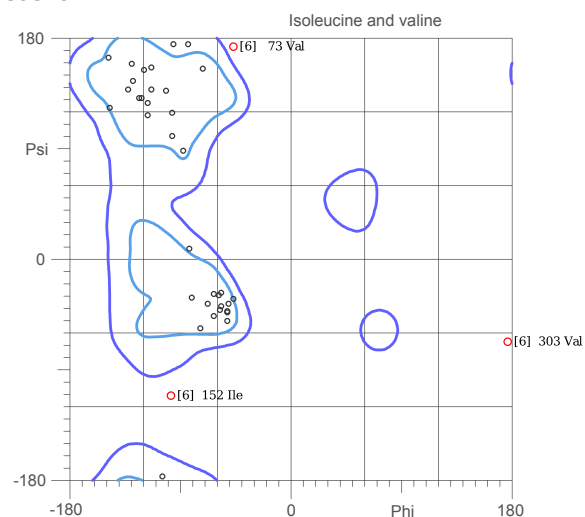
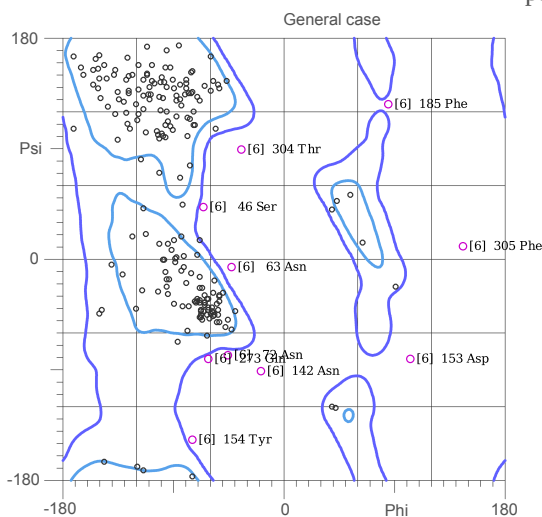
88.5% (269/304) of all residues were in favored (98%) regions.
95.7% (291/304) of all residues were in allowed (>99.8%) regions.

There were 13 outliers (phi, psi):

[5] 63 Asn (-52.4, 45.8)	[5] 221 Asn (-174.7, 74.7)
[5] 73 Val (-150.2, 46.8)	[5] 222 Arg (-1.1, 74.8)
[5] 74 Gln (-8.1, 92.2)	[5] 301 Ser (-160.1, -137.5)
[5] 153 Asp (-18.7, -54.1)	[5] 302 Gly (-46.9, 85.3)
[5] 154 Tyr (-7.6, -42.1)	[5] 303 Val (-72.3, 21.2)
[5] 155 Asp (112.8, -9.4)	
[5] 184 Pro (-69.7, -90.0)	
[5] 185 Phe (63.1, 124.0)	

MolProbity Ramachandran analysis

7mhp.H.pdb, model 6



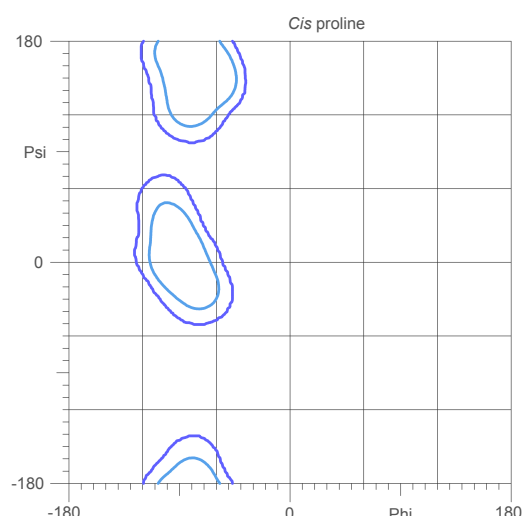
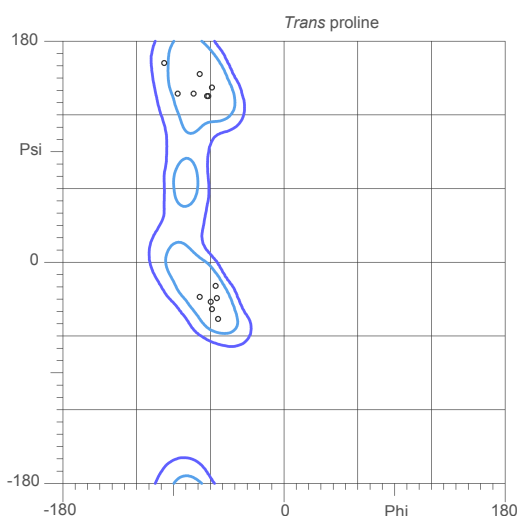
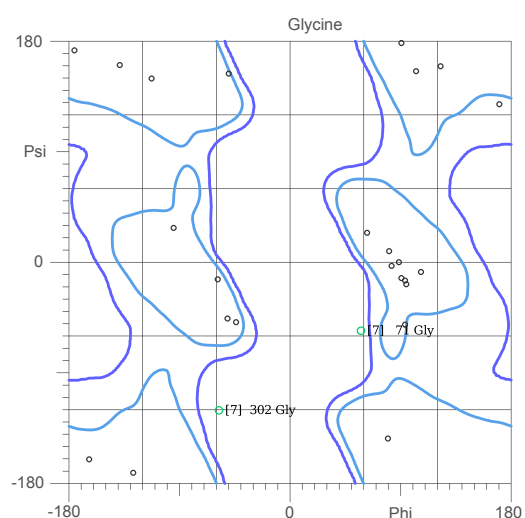
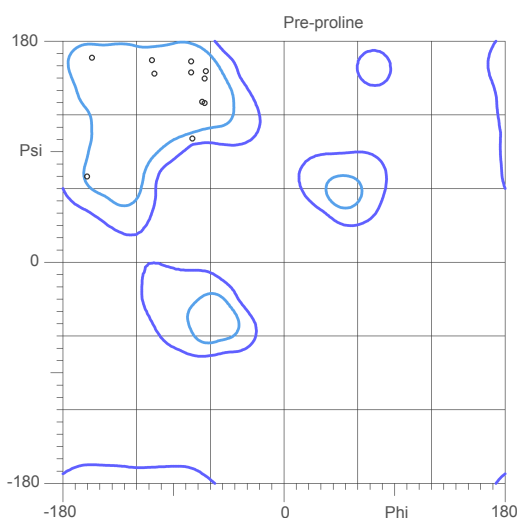
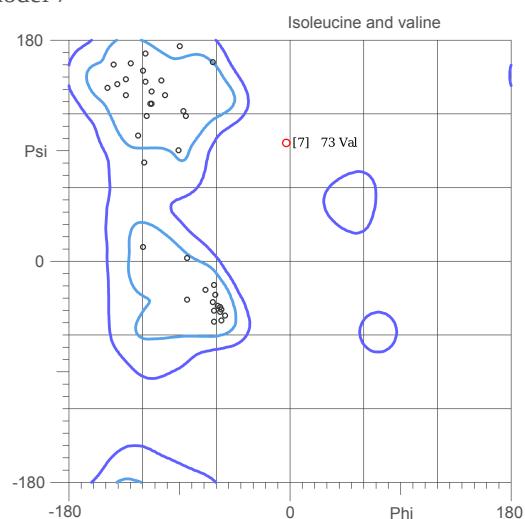
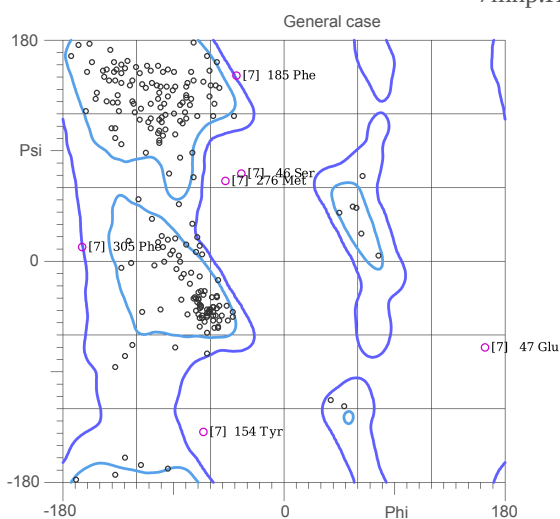
86.8% (264/304) of all residues were in favored (98%) regions.
94.7% (288/304) of all residues were in allowed (>99.8%) regions.

There were 16 outliers (phi, psi):

[6] 46 Ser (-66.9, 43.2)	[6] 184 Pro (-69.3, -138.8)
[6] 63 Asn (-43.3, -6.2)	[6] 185 Phe (85.6, 127.3)
[6] 72 Asn (-46.5, -79.0)	[6] 195 Gly (-51.1, -172.7)
[6] 73 Val (-47.5, 174.4)	[6] 273 Gln (-62.6, -81.0)
[6] 142 Asn (-19.7, -92.0)	[6] 302 Gly (-49.2, -117.1)
[6] 152 Ile (-98.3, -111.2)	[6] 303 Val (177.3, -67.5)
[6] 153 Asp (103.4, -81.7)	[6] 304 Thr (-35.9, 90.7)
[6] 154 Tyr (-75.7, -147.6)	[6] 305 Phe (146.4, 11.6)

MolProbity Ramachandran analysis

7mhp.H.pdb, model 7



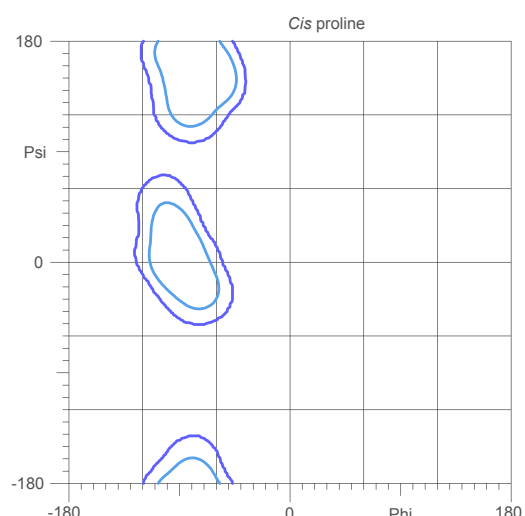
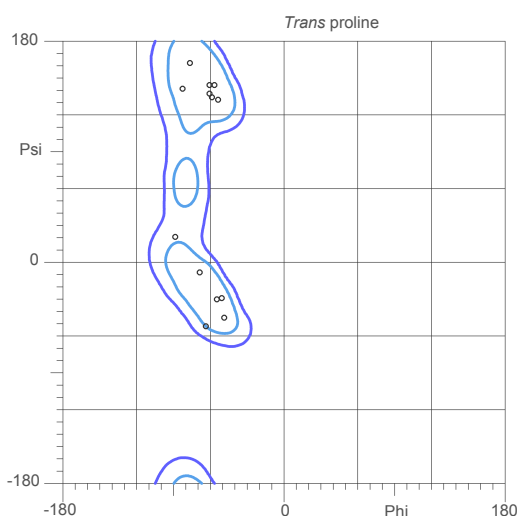
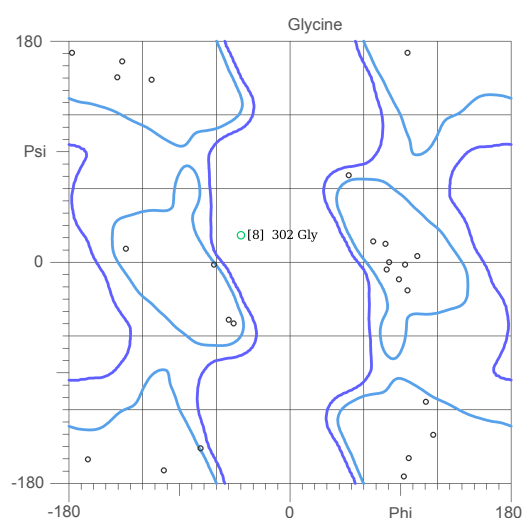
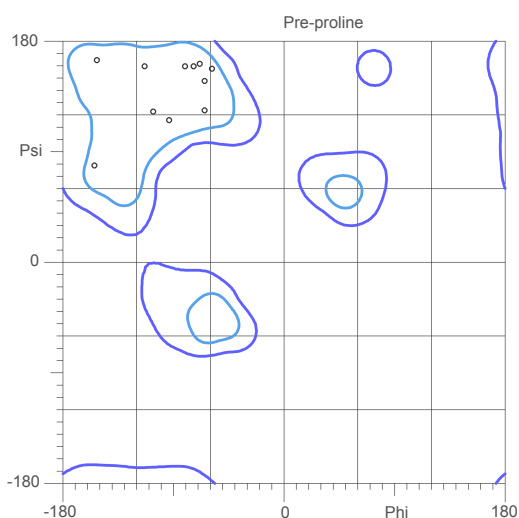
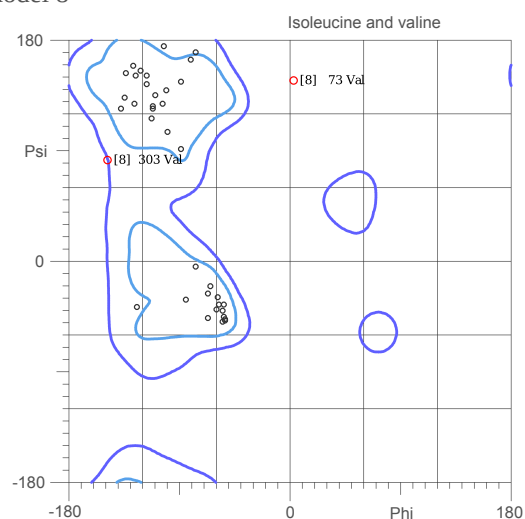
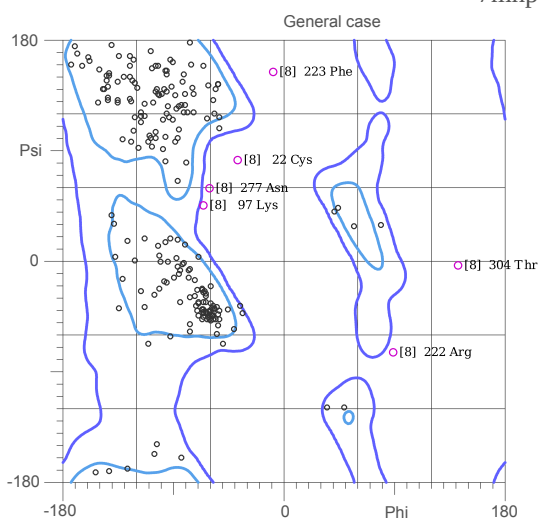
87.5% (266/304) of all residues were in favored (98%) regions.
97.0% (295/304) of all residues were in allowed (>99.8%) regions.

There were 9 outliers (phi, psi):

[7] 46 Ser (-35.3, 72.5)	[7] 305 Phe (-165.2, 12.1)
[7] 47 Glu (164.6, -70.7)	
[7] 71 Gly (58.0, -56.4)	
[7] 73 Val (-3.0, 97.1)	
[7] 154 Tyr (-66.9, -139.9)	
[7] 185 Phe (-39.0, 152.1)	
[7] 276 Met (-48.5, 66.4)	
[7] 302 Gly (-58.4, -121.8)	

MolProbity Ramachandran analysis

7mhp.H.pdb, model 8



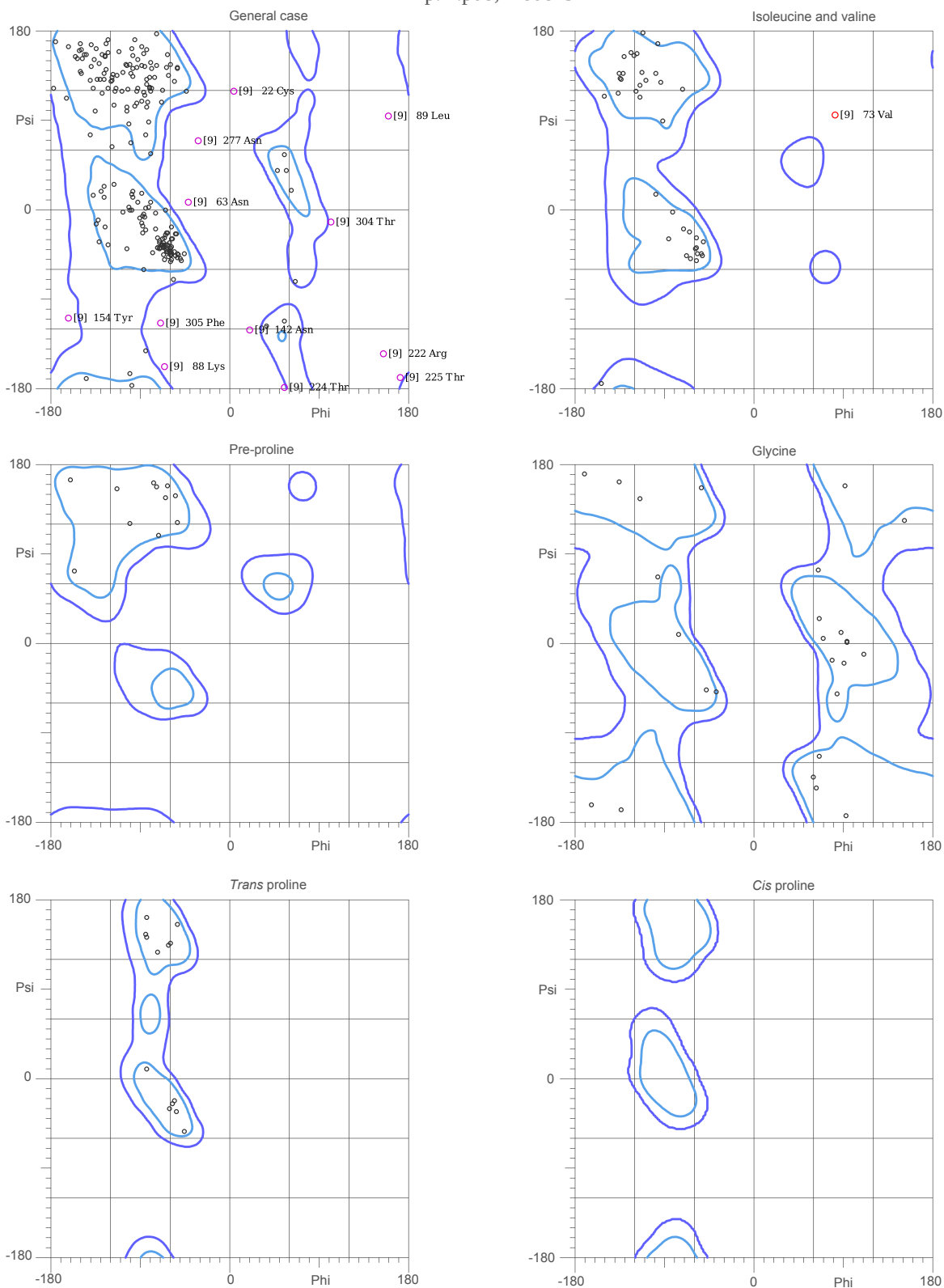
90.1% (274/304) of all residues were in favored (98%) regions.
97.0% (295/304) of all residues were in allowed (>99.8%) regions.

There were 9 outliers (phi, psi):

[8] 22 Cys (-38.3, 83.2)	[8] 304 Thr (142.1, -3.9)
[8] 73 Val (3.7, 148.2)	
[8] 97 Lys (-66.2, 46.3)	
[8] 222 Arg (89.3, -74.6)	
[8] 223 Phe (-10.0, 155.6)	
[8] 277 Asn (-61.0, 60.5)	
[8] 302 Gly (-40.2, 22.5)	
[8] 303 Val (-149.0, 83.6)	

MolProbity Ramachandran analysis

7mhp.H.pdb, model 9



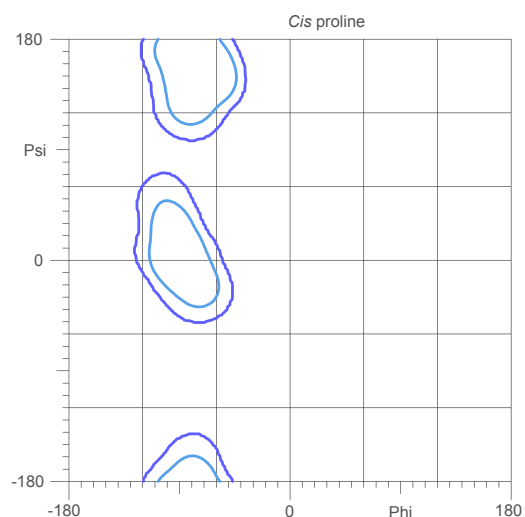
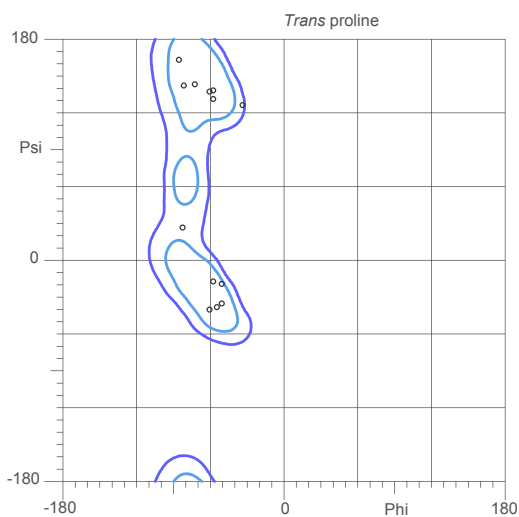
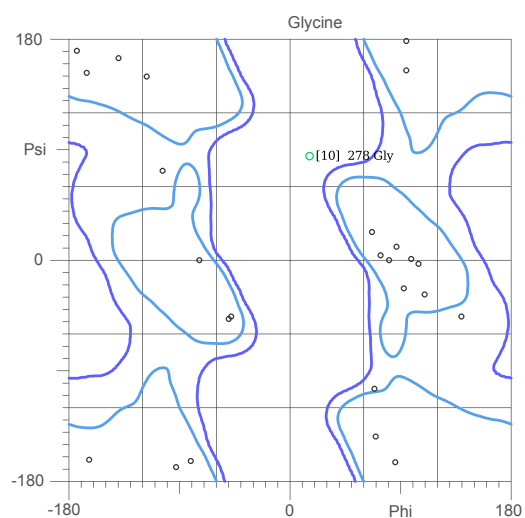
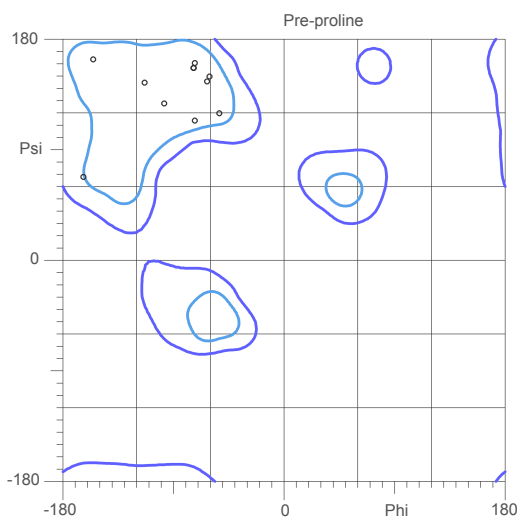
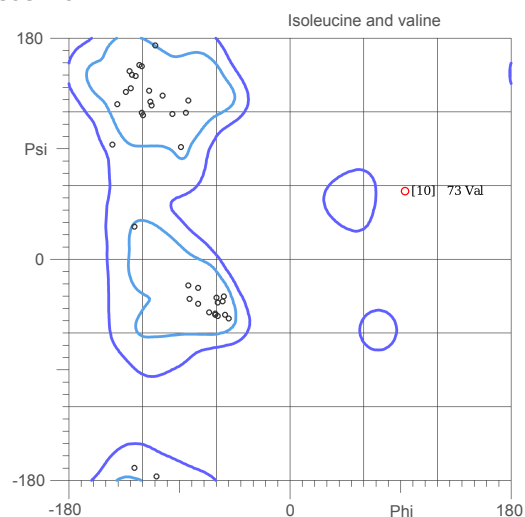
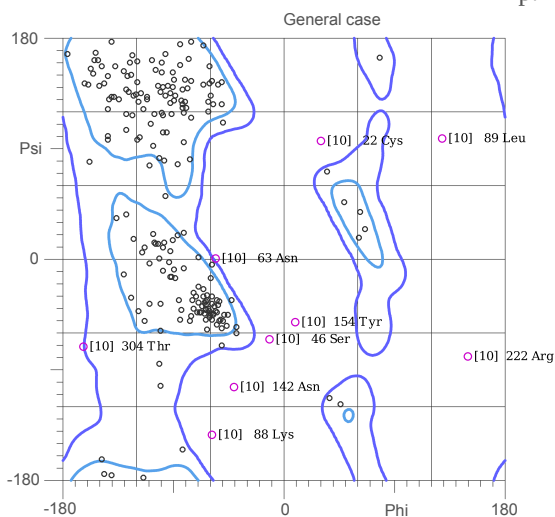
88.8% (270/304) of all residues were in favored (98%) regions.
95.7% (291/304) of all residues were in allowed (>99.8%) regions.

There were 13 outliers (phi, psi):

[9] 22 Cys (4.8, 120.4)	[9] 224 Thr (55.7, -179.1)
[9] 63 Asn (-42.9, 8.5)	[9] 225 Thr (172.9, -169.2)
[9] 73 Val (82.5, 97.0)	[9] 277 Asn (-32.6, 70.4)
[9] 88 Lys (-66.6, -158.0)	[9] 304 Thr (102.0, -12.5)
[9] 89 Leu (160.4, 95.4)	[9] 305 Phe (-70.5, -114.3)
[9] 142 Asn (20.6, -121.2)	
[9] 154 Tyr (-164.0, -109.3)	
[9] 222 Arg (155.7, -145.3)	

MolProbity Ramachandran analysis

7mhp.H.pdb, model 10



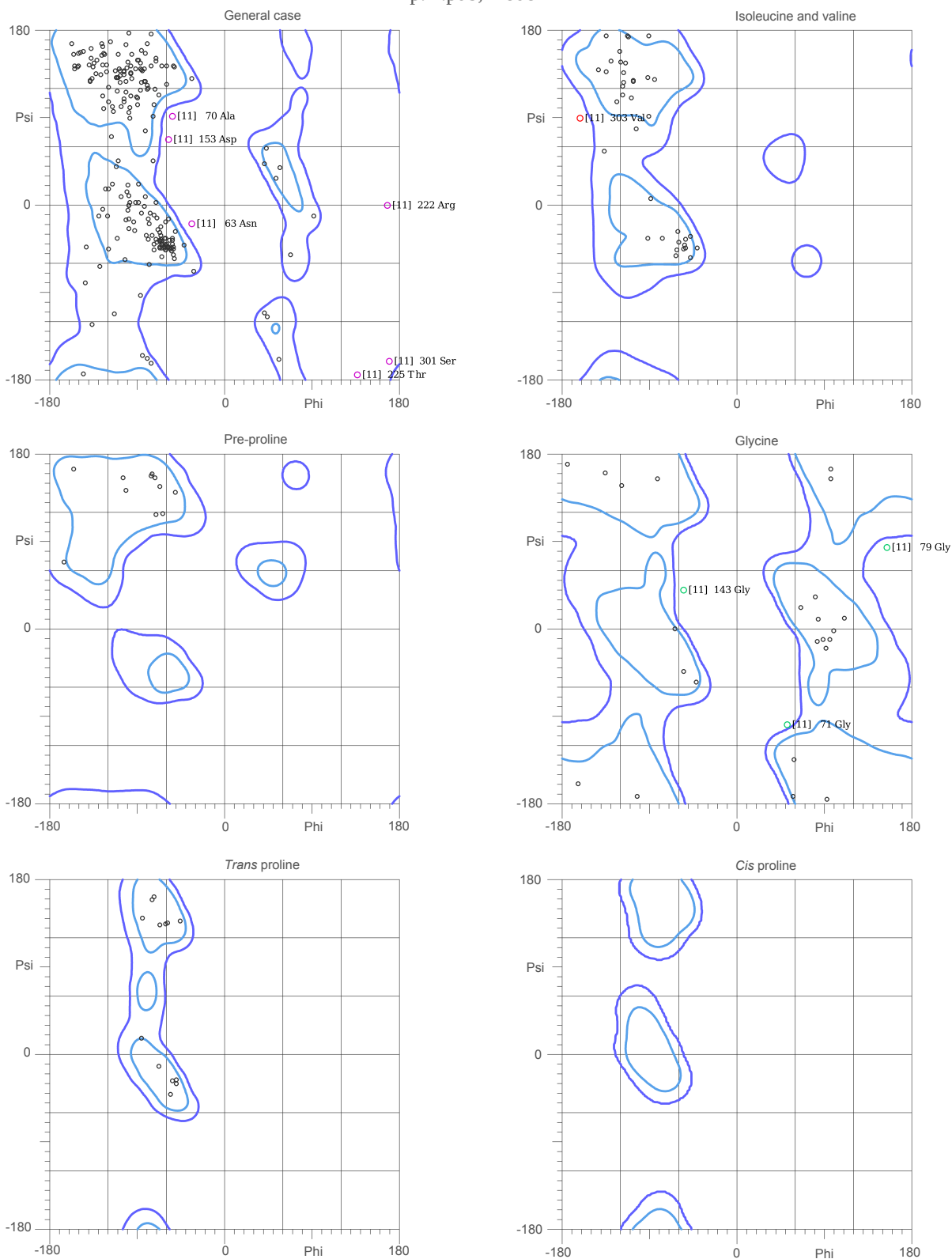
86.8% (264/304) of all residues were in favored (98%) regions.
96.4% (293/304) of all residues were in allowed (>99.8%) regions.

There were 11 outliers (phi, psi):

[10] 22 Cys (30.5, 97.9)	[10] 222 Arg (151.0, -79.9)
[10] 46 Ser (-12.3, -65.1)	[10] 278 Gly (16.2, 85.7)
[10] 63 Asn (-56.4, 2.0)	[10] 304 Thr (-165.0, -71.2)
[10] 73 Val (94.0, 56.1)	
[10] 88 Lys (-59.6, -143.9)	
[10] 89 Leu (129.2, 99.5)	
[10] 142 Asn (-41.7, -104.8)	
[10] 154 Tyr (9.6, -51.5)	

MolProbity Ramachandran analysis

7mhp.H.pdb, model 11



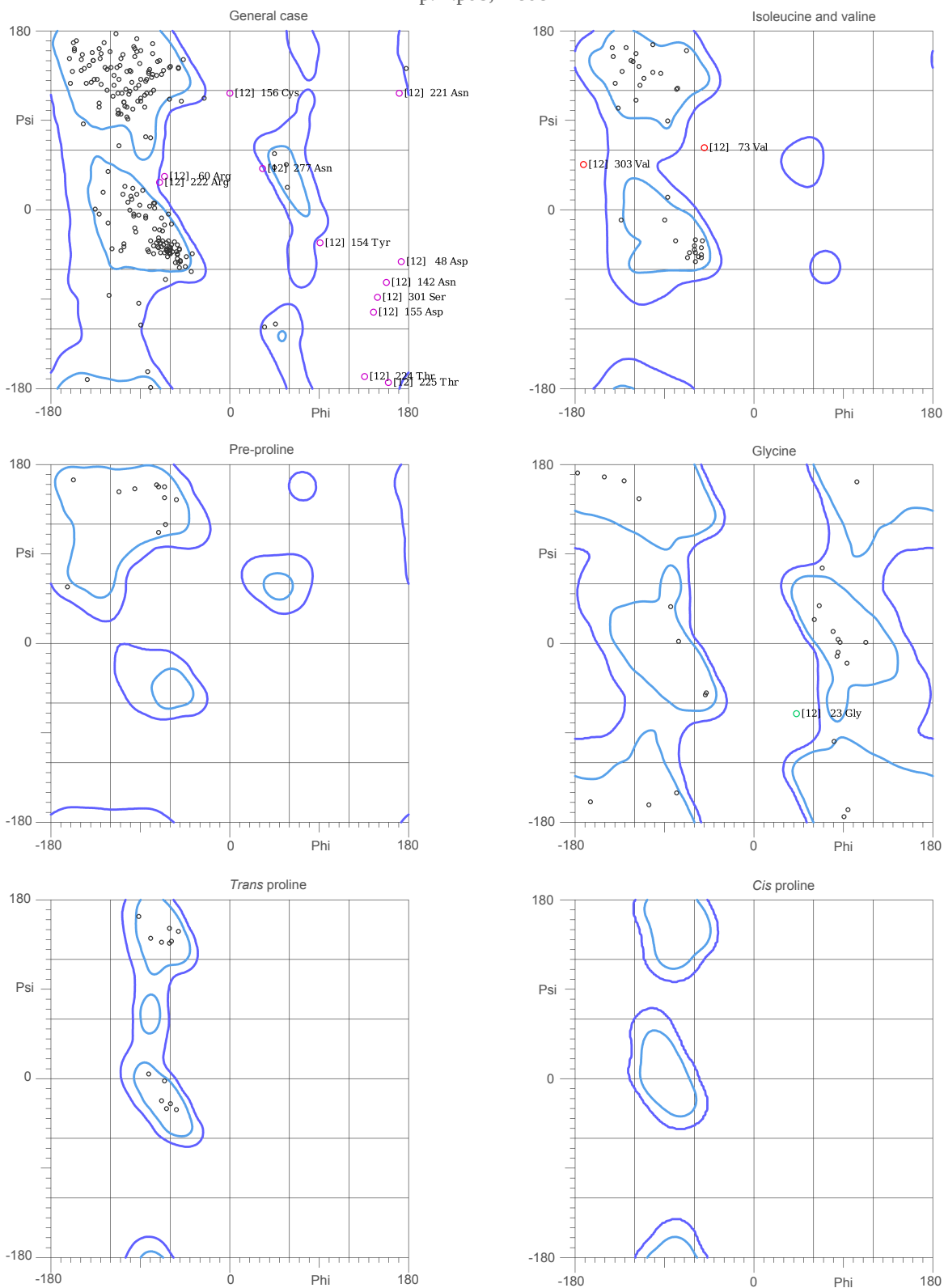
88.5% (269/304) of all residues were in favored (98%) regions.
96.7% (294/304) of all residues were in allowed (>99.8%) regions.

There were 10 outliers (phi, psi):

[11] 63 Asn (-34.5, -19.2)	[11] 301 Ser (170.5, -161.0)
[11] 70 Ala (-54.3, 92.1)	[11] 303 Val (-162.3, 90.7)
[11] 71 Gly (52.3, -99.5)	
[11] 79 Gly (155.0, 84.4)	
[11] 143 Gly (-55.7, 40.6)	
[11] 153 Asp (-59.0, 68.1)	
[11] 222 Arg (168.7, -0.6)	
[11] 225 Thr (137.9, -175.2)	

MolProbity Ramachandran analysis

7mhp.H.pdb, model 12



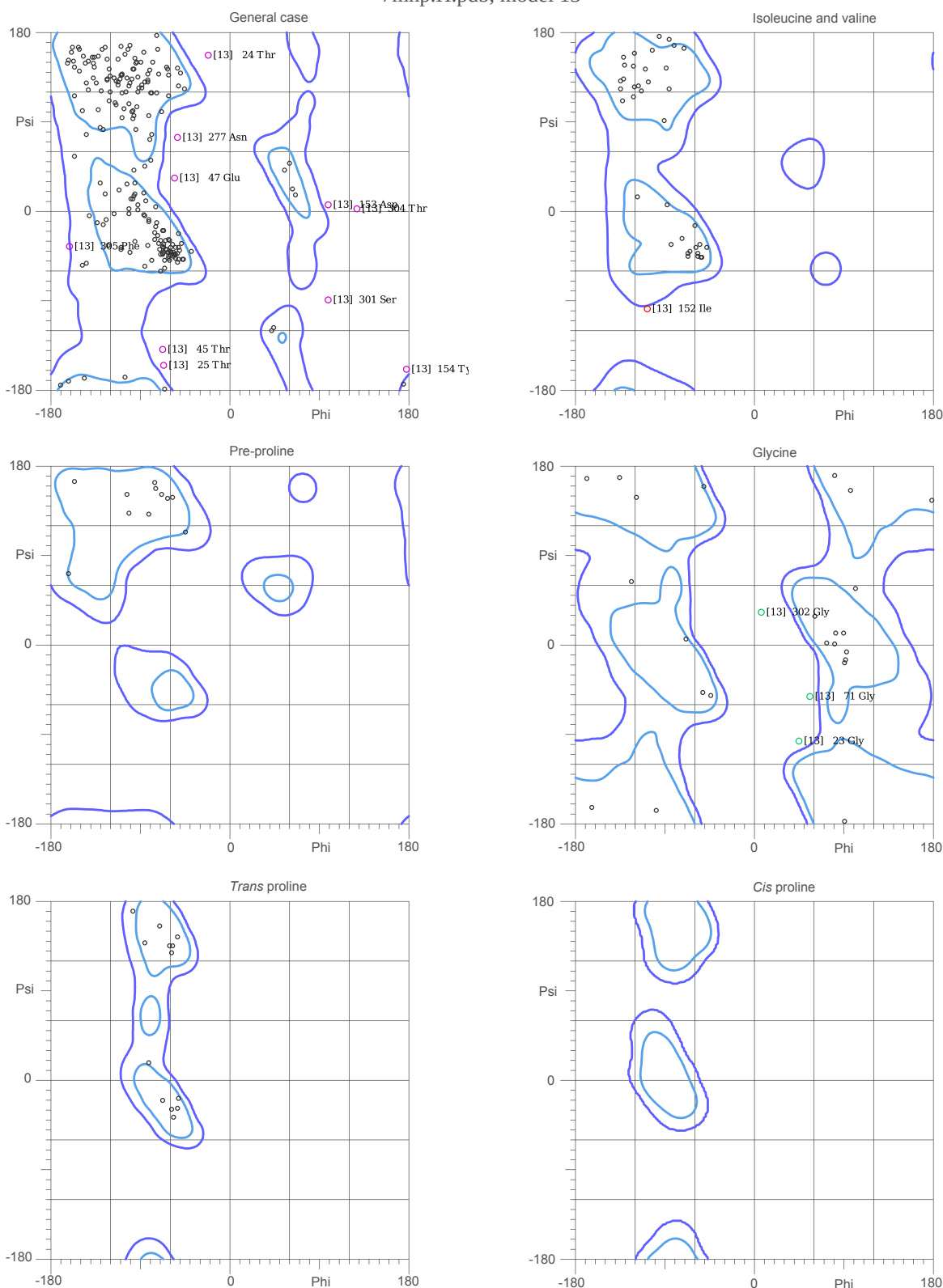
88.2% (268/304) of all residues were in favored (98%) regions.
95.1% (289/304) of all residues were in allowed (>99.8%) regions.

There were 15 outliers (phi, psi):

[12] 23 Gly (43.9, -71.8)	[12] 221 Asn (171.9, 118.5)
[12] 48 Asp (173.2, -52.3)	[12] 222 Arg (-71.1, 28.6)
[12] 60 Arg (-66.9, 34.1)	[12] 224 Thr (136.2, -168.9)
[12] 73 Val (-50.3, 63.6)	[12] 225 Thr (161.0, -174.8)
[12] 142 Asn (158.3, -73.4)	[12] 277 Asn (33.6, 42.9)
[12] 154 Tyr (91.2, -33.3)	[12] 301 Ser (149.0, -89.0)
[12] 155 Asp (145.5, -103.4)	[12] 303 Val (-172.1, 46.4)
[12] 156 Cys (0.7, 118.3)	

MolProbity Ramachandran analysis

7mhp.H.pdb, model 13



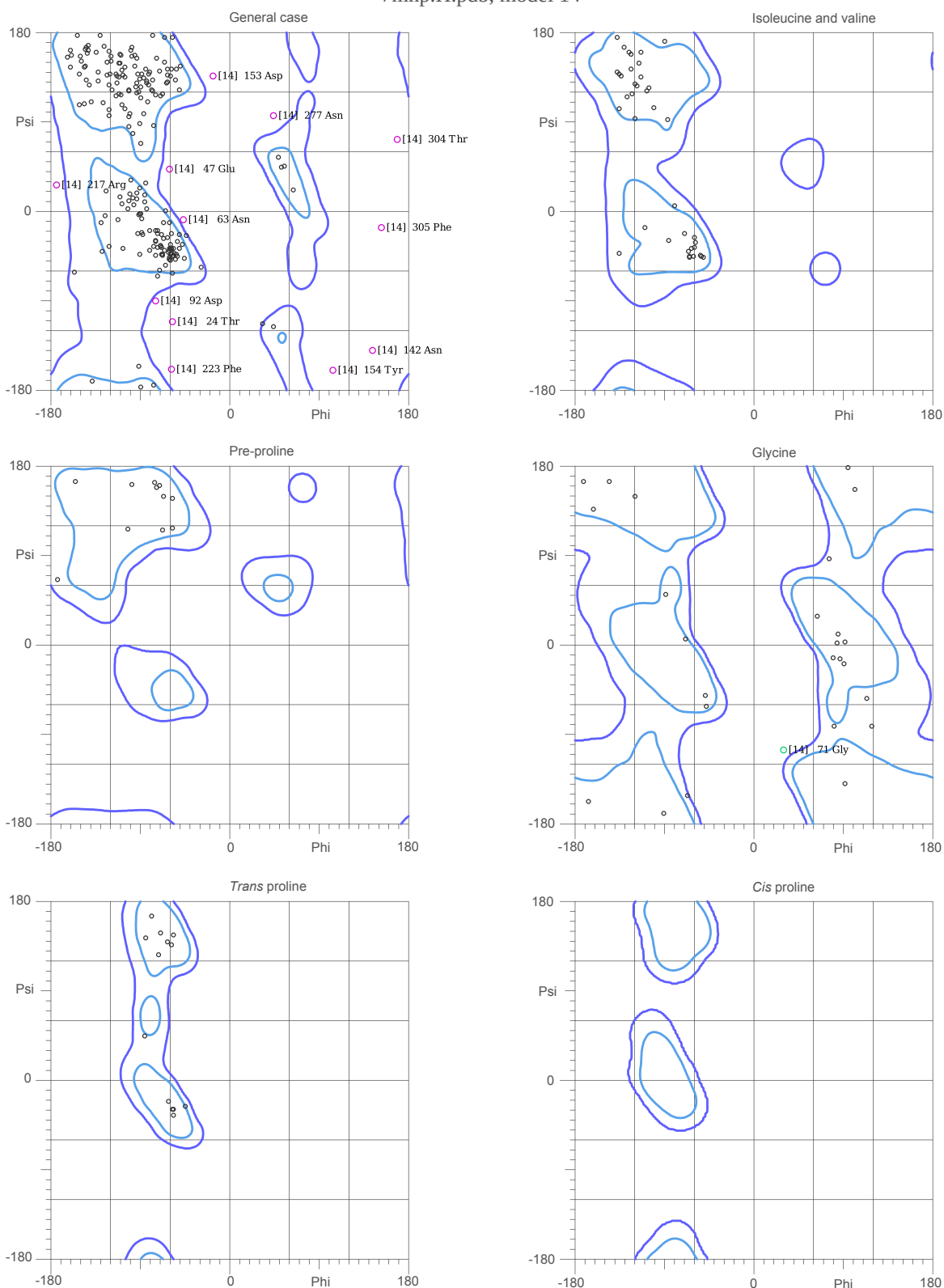
86.8% (264/304) of all residues were in favored (98%) regions.
95.4% (290/304) of all residues were in allowed (>99.8%) regions.

There were 14 outliers (phi, psi):

[13] 23 Gly (45.7, -97.2)	[13] 154 Tyr (178.3, -159.2)
[13] 24 Thr (-22.1, 158.8)	[13] 277 Asn (-53.1, 75.1)
[13] 25 Thr (-67.1, -155.6)	[13] 301 Ser (100.0, -89.5)
[13] 45 Thr (-68.8, -139.4)	[13] 302 Gly (7.4, 33.6)
[13] 47 Glu (-56.4, 34.1)	[13] 304 Thr (128.3, 3.7)
[13] 71 Gly (56.1, -52.6)	[13] 305 Phe (-162.8, -35.6)
[13] 152 Ile (-108.5, -98.9)	
[13] 153 Asp (99.9, 7.2)	

MolProbity Ramachandran analysis

7mhp.H.pdb, model 14



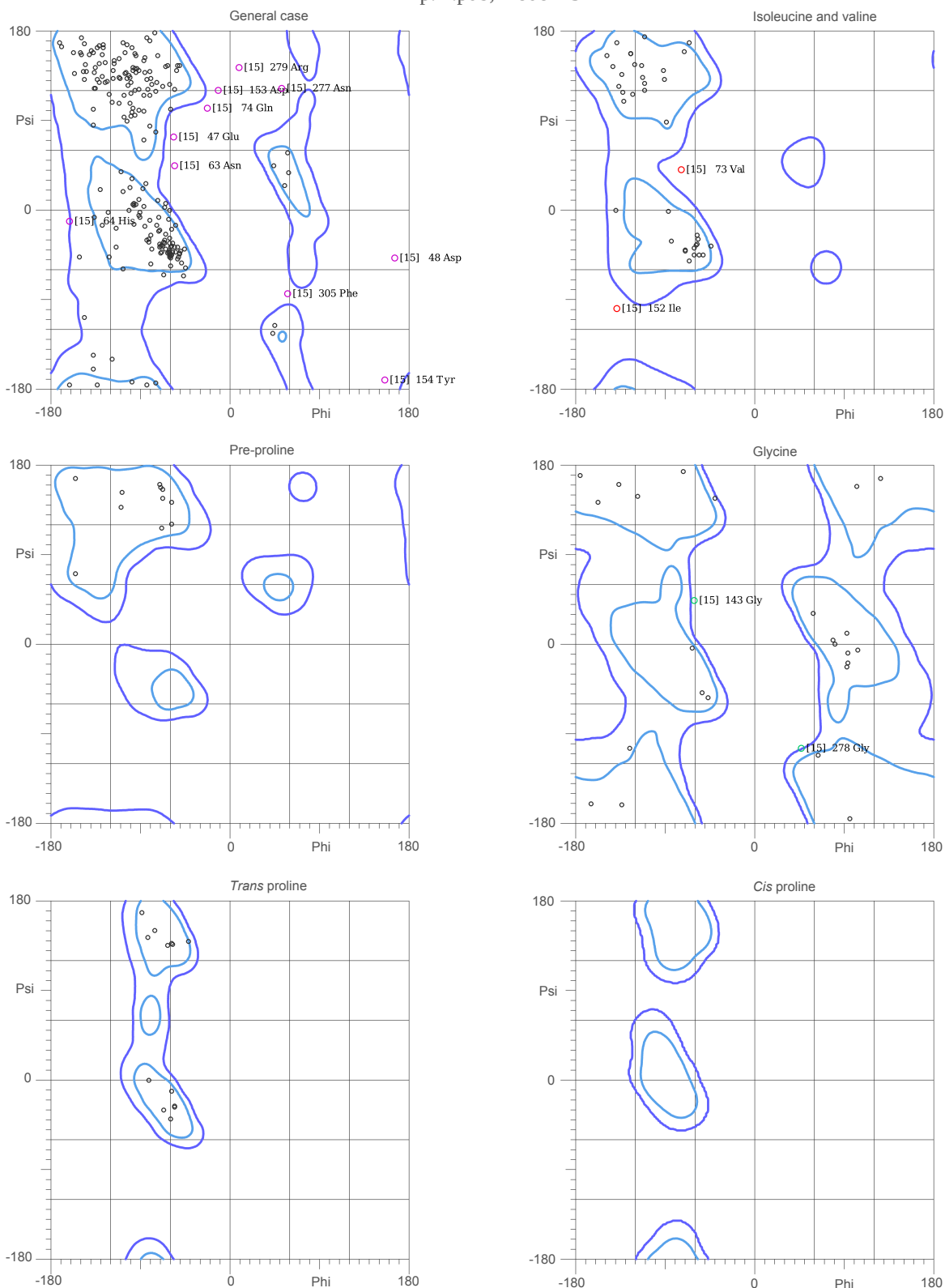
88.2% (268/304) of all residues were in favored (98%) regions.
95.7% (291/304) of all residues were in allowed (>99.8%) regions.

There were 13 outliers (phi, psi):

[14] 24 Thr (-58.0, -111.6)	[14] 217 Arg (-175.8, 27.5)
[14] 47 Glu (-61.9, 43.2)	[14] 223 Phe (-59.3, -159.6)
[14] 63 Asn (-47.1, -8.7)	[14] 277 Asn (44.7, 97.6)
[14] 71 Gly (30.2, -106.9)	[14] 304 Thr (169.6, 73.6)
[14] 92 Asp (-75.1, -90.5)	[14] 305 Phe (153.5, -16.6)
[14] 142 Asn (145.0, -140.7)	
[14] 153 Asp (-17.4, 138.0)	
[14] 154 Tyr (104.2, -160.3)	

MolProbity Ramachandran analysis

7mhp.H.pdb, model 15



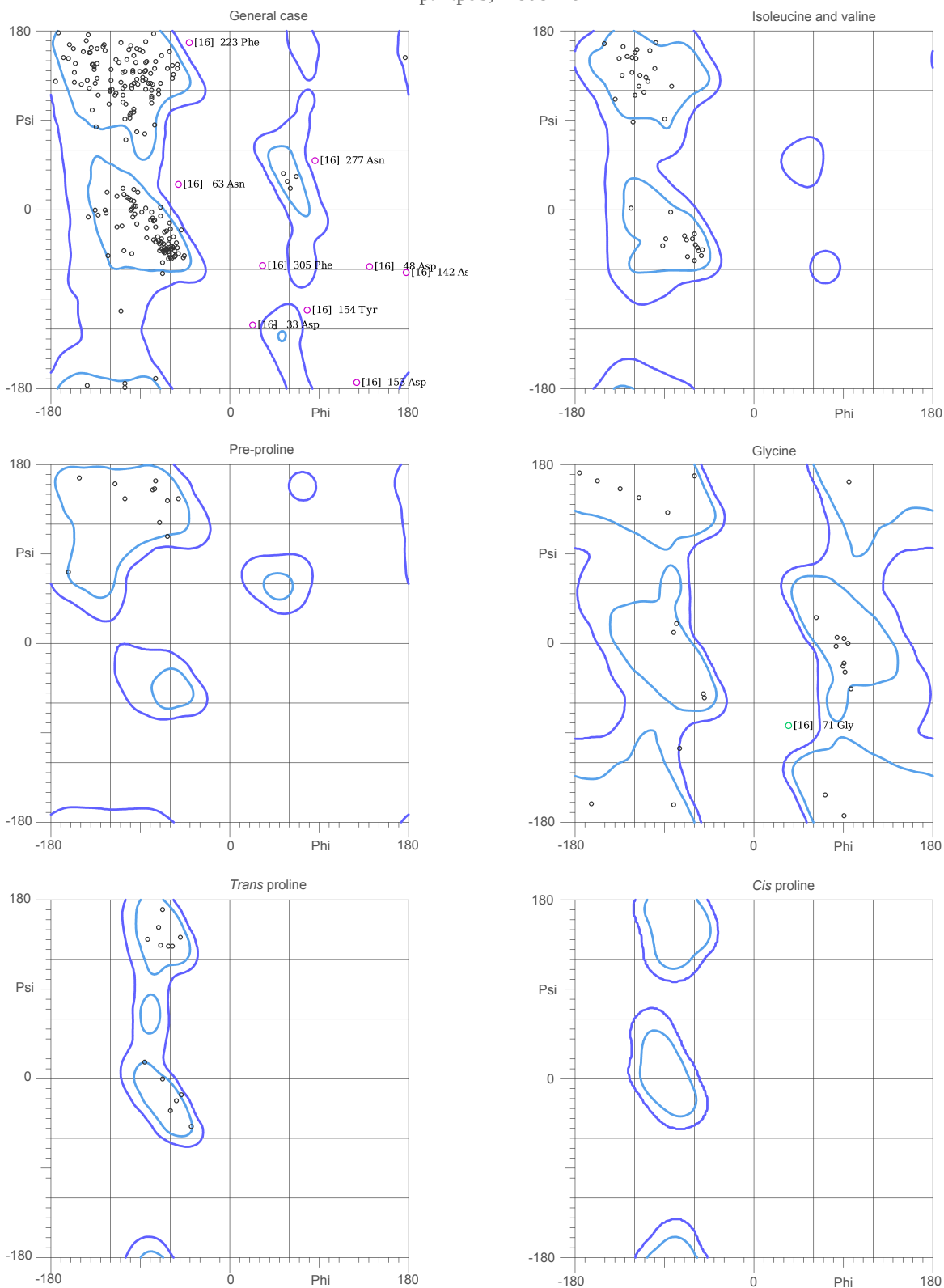
87.5% (266/304) of all residues were in favored (98%) regions.
95.4% (290/304) of all residues were in allowed (>99.8%) regions.

There were 14 outliers (phi, psi):

[15] 47 Glu (-57.9, 74.7)	[15] 153 Asp (-13.0, 121.7)
[15] 48 Asp (166.6, -48.4)	[15] 154 Tyr (156.8, -171.0)
[15] 63 Asn (-56.5, 45.8)	[15] 277 Asn (52.5, 123.4)
[15] 64 His (-162.2, -11.4)	[15] 278 Gly (47.5, -105.1)
[15] 73 Val (-74.3, 41.0)	[15] 279 Arg (9.1, 144.0)
[15] 74 Gln (-23.5, 103.1)	[15] 305 Phe (58.6, -84.3)
[15] 143 Gly (-61.7, 44.1)	
[15] 152 Ile (-139.1, -99.5)	

MolProbity Ramachandran analysis

7mhp.H.pdb, model 16



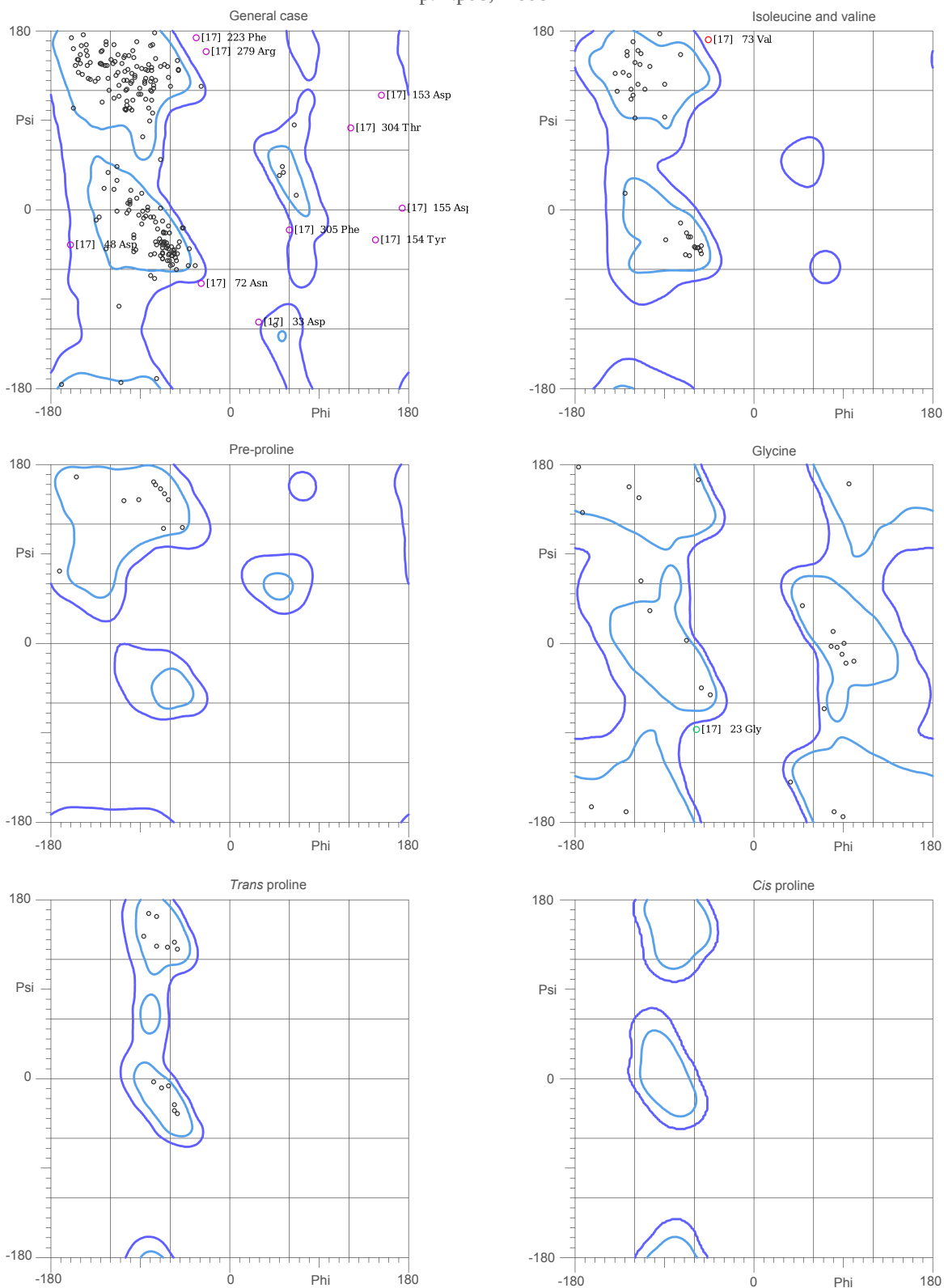
90.5% (275/304) of all residues were in favored (98%) regions.
96.7% (294/304) of all residues were in allowed (>99.8%) regions.

There were 10 outliers (phi, psi):

[16] 33 Asp (23.2, -116.1)	[16] 277 Asn (86.7, 50.8)
[16] 48 Asp (141.4, -57.1)	[16] 305 Phe (33.4, -56.8)
[16] 63 Asn (-52.2, 26.1)	
[16] 71 Gly (36.0, -83.2)	
[16] 142 Asn (178.6, -63.8)	
[16] 153 Asp (128.9, -174.6)	
[16] 154 Tyr (78.1, -101.2)	
[16] 223 Phe (-41.4, 169.3)	

MolProbity Ramachandran analysis

7mhp.H.pdb, model 17



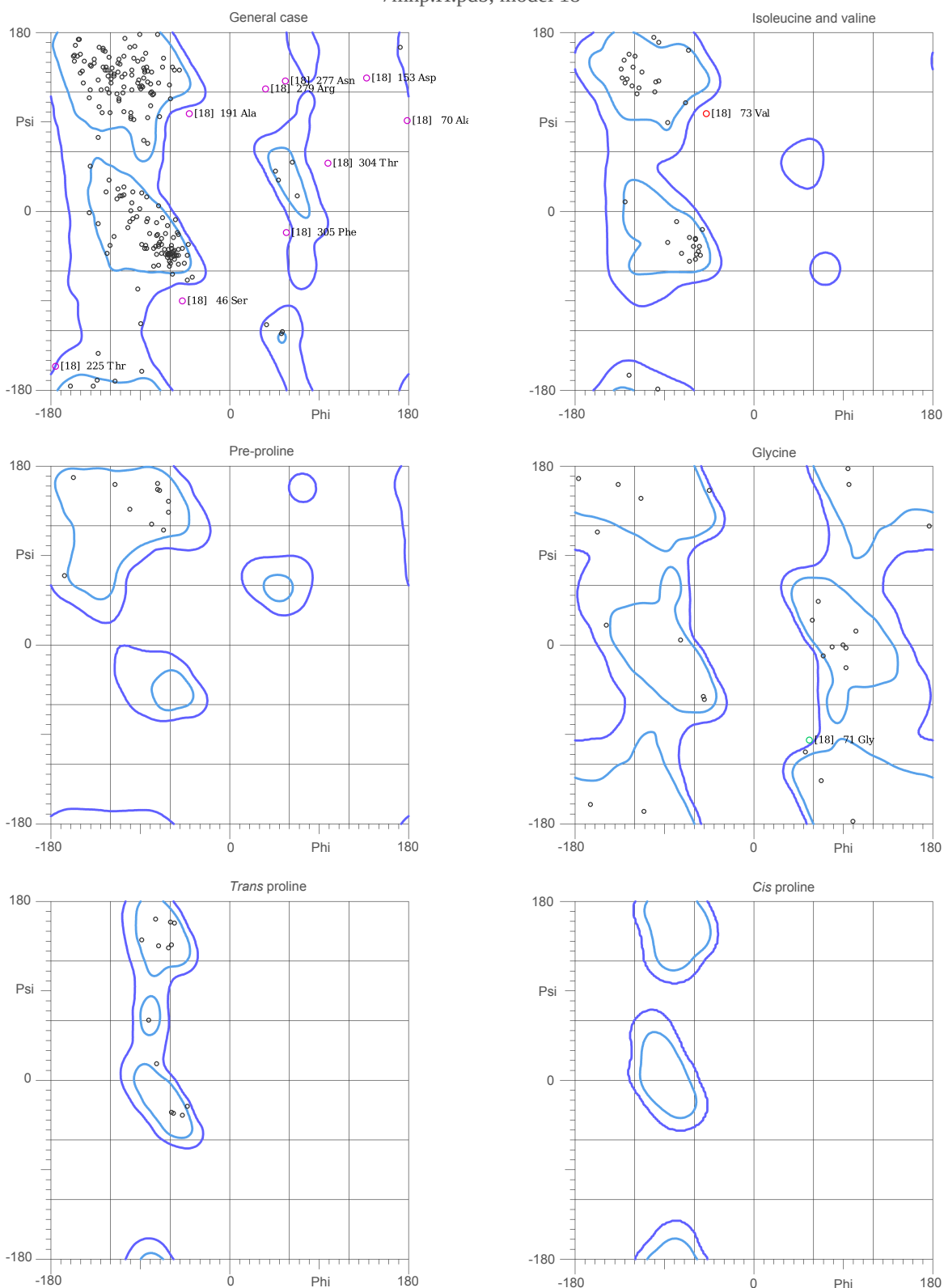
89.5% (272/304) of all residues were in favored (98%) regions.
96.1% (292/304) of all residues were in allowed (>99.8%) regions.

There were 12 outliers (phi, psi):

[17] 23 Gly (-58.5, -87.2)	[17] 223 Phe (-34.0, 174.4)
[17] 33 Asp (29.2, -113.9)	[17] 279 Arg (-24.5, 160.7)
[17] 48 Asp (-161.2, -35.3)	[17] 304 Thr (122.4, 83.8)
[17] 72 Asn (-29.7, -74.8)	[17] 305 Phe (60.6, -20.3)
[17] 73 Val (-46.8, 172.9)	
[17] 153 Asp (153.4, 116.4)	
[17] 154 Tyr (147.1, -30.6)	
[17] 155 Asp (174.9, 2.1)	

MolProbity Ramachandran analysis

7mhp.H.pdb, model 18



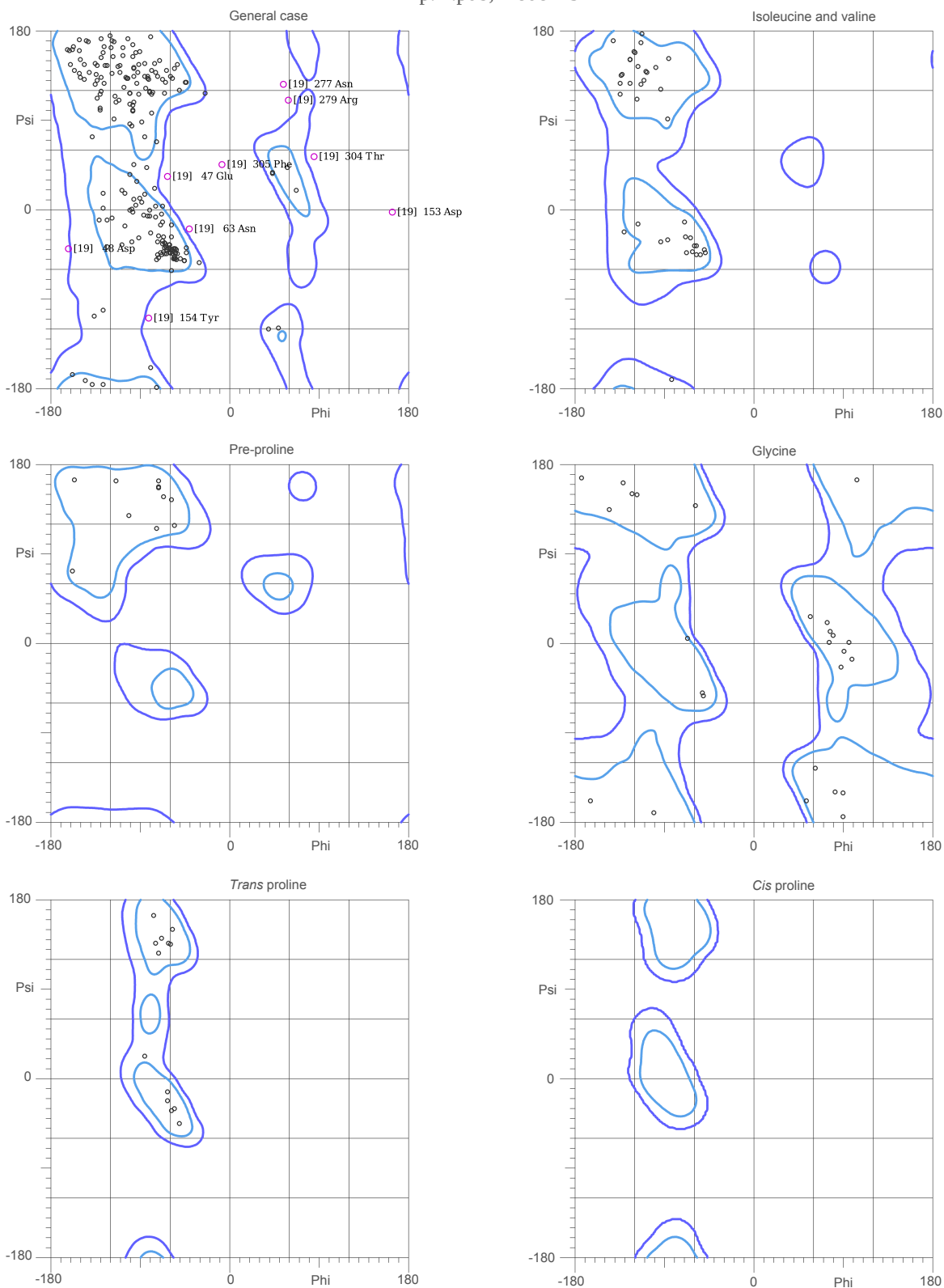
86.8% (264/304) of all residues were in favored (98%) regions.
96.4% (293/304) of all residues were in allowed (>99.8%) regions.

There were 11 outliers (phi, psi):

[18] 46 Ser (-48.5, -90.5)	[18] 279 Arg (36.3, 124.1)
[18] 70 Ala (179.7, 92.7)	[18] 304 Thr (99.4, 49.7)
[18] 71 Gly (56.9, -96.2)	[18] 305 Phe (57.9, -21.3)
[18] 73 Val (-48.2, 100.0)	
[18] 153 Asp (138.6, 136.0)	
[18] 191 Ala (-41.7, 99.6)	
[18] 225 Thr (-176.0, -156.5)	
[18] 277 Asn (56.1, 132.6)	

MolProbity Ramachandran analysis

7mhp.H.pdb, model 19



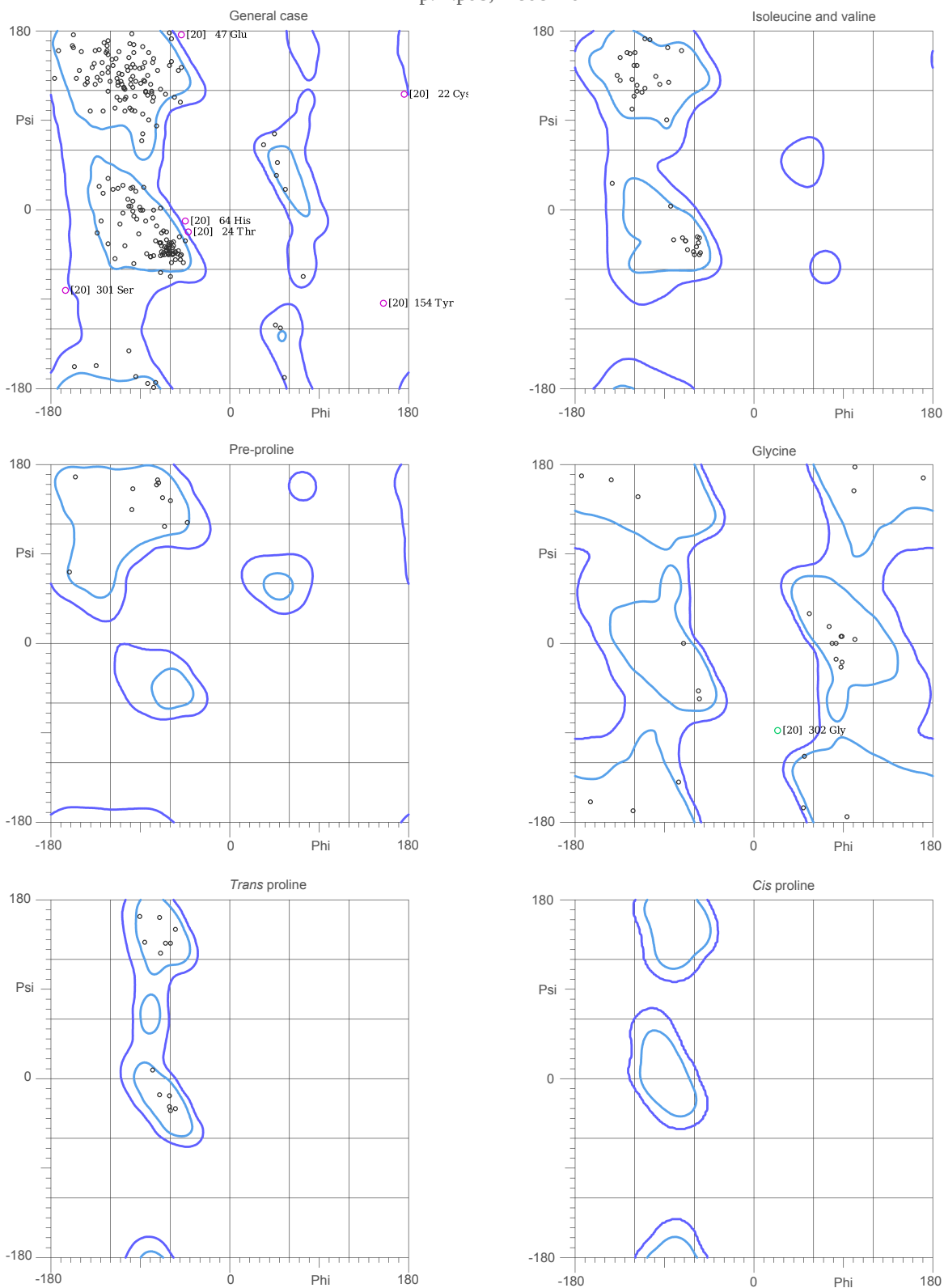
90.5% (275/304) of all residues were in favored (98%) regions.
97.0% (295/304) of all residues were in allowed (>99.8%) regions.

There were 9 outliers (phi, psi):

[19] 47 Glu (-63.4, 34.6)	[19] 305 Phe (-8.2, 47.0)
[19] 48 Asp (-163.3, -39.5)	
[19] 63 Asn (-41.2, -19.8)	
[19] 153 Asp (164.4, -3.0)	
[19] 154 Tyr (-82.0, -109.3)	
[19] 277 Asn (54.4, 127.2)	
[19] 279 Arg (59.7, 111.0)	
[19] 304 Thr (85.3, 54.8)	

MolProbity Ramachandran analysis

7mhp.H.pdb, model 20



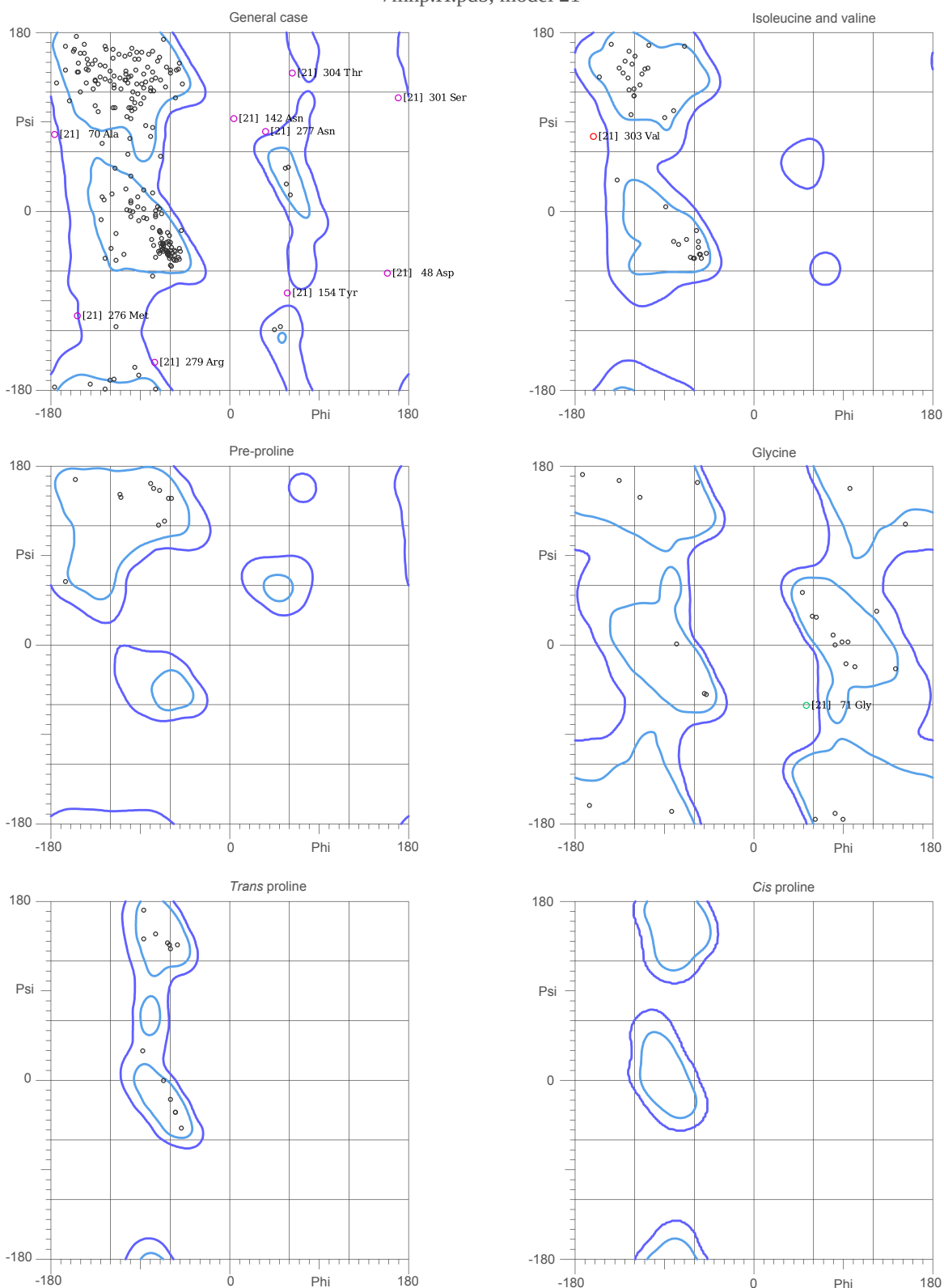
89.8% (273/304) of all residues were in favored (98%) regions.
97.7% (297/304) of all residues were in allowed (>99.8%) regions.

There were 7 outliers (phi, psi):

[20] 22 Cys (176.7, 117.3)
[20] 24 Thr (-42.8, -22.1)
[20] 47 Glu (-49.0, 177.4)
[20] 64 His (-46.0, -11.2)
[20] 154 Tyr (155.8, -94.1)
[20] 301 Ser (-166.1, -81.3)
[20] 302 Gly (24.7, -88.9)

MolProbity Ramachandran analysis

7mhp.H.pdb, model 21



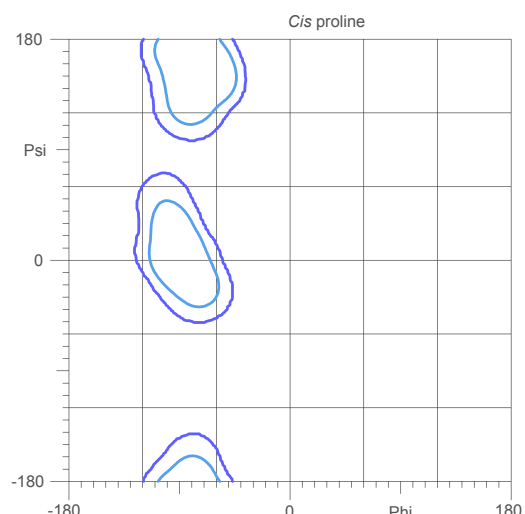
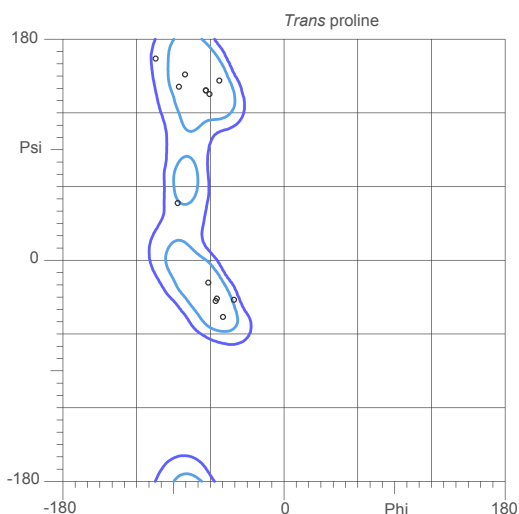
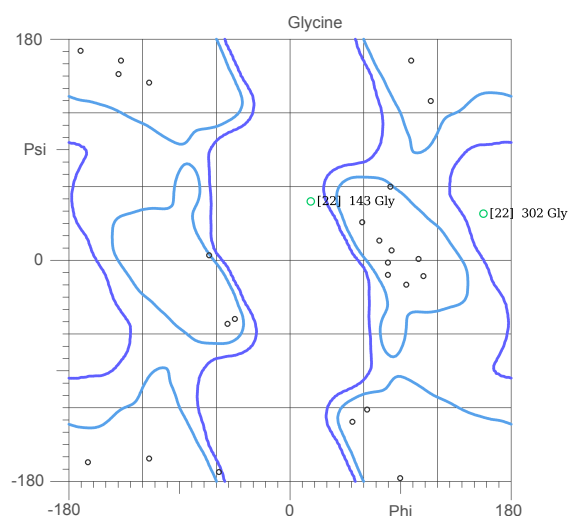
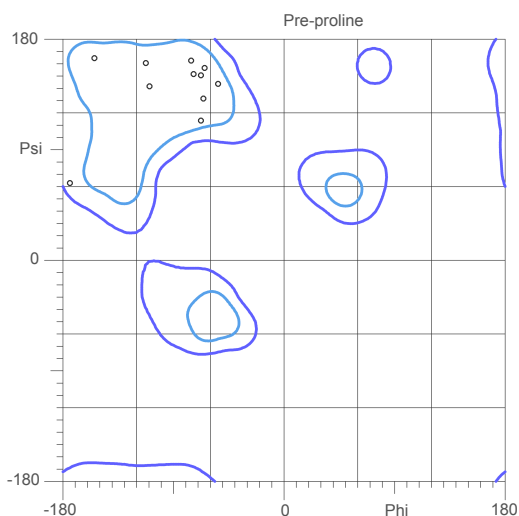
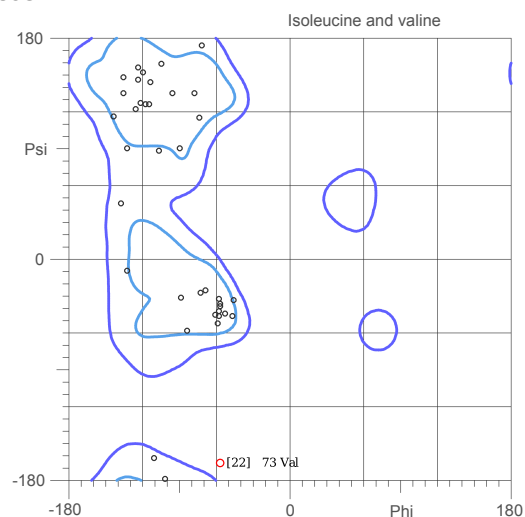
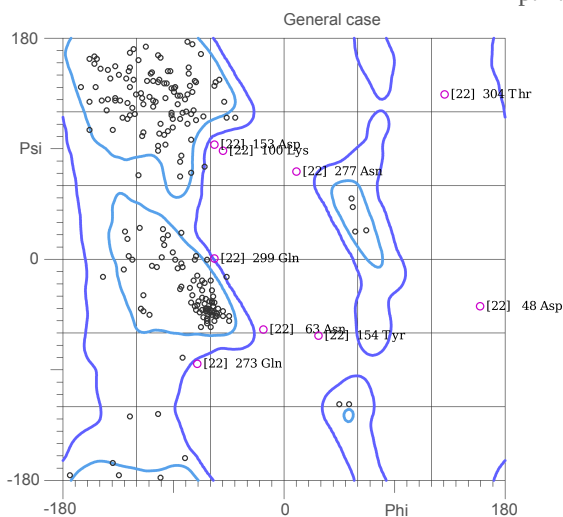
88.2% (268/304) of all residues were in favored (98%) regions.
96.4% (293/304) of all residues were in allowed (>99.8%) regions.

There were 11 outliers (phi, psi):

[21] 48 Asp (159.6, -62.7)	[21] 301 Ser (170.9, 115.7)
[21] 70 Ala (-177.8, 78.3)	[21] 303 Val (-162.4, 76.7)
[21] 71 Gly (53.1, -61.8)	[21] 304 Thr (63.3, 140.5)
[21] 142 Asn (4.7, 94.1)	
[21] 154 Tyr (58.3, -82.5)	
[21] 276 Met (-154.9, -105.2)	
[21] 277 Asn (36.9, 81.5)	
[21] 279 Arg (-76.8, -152.7)	

MolProbity Ramachandran analysis

7mhp.H.pdb, model 22



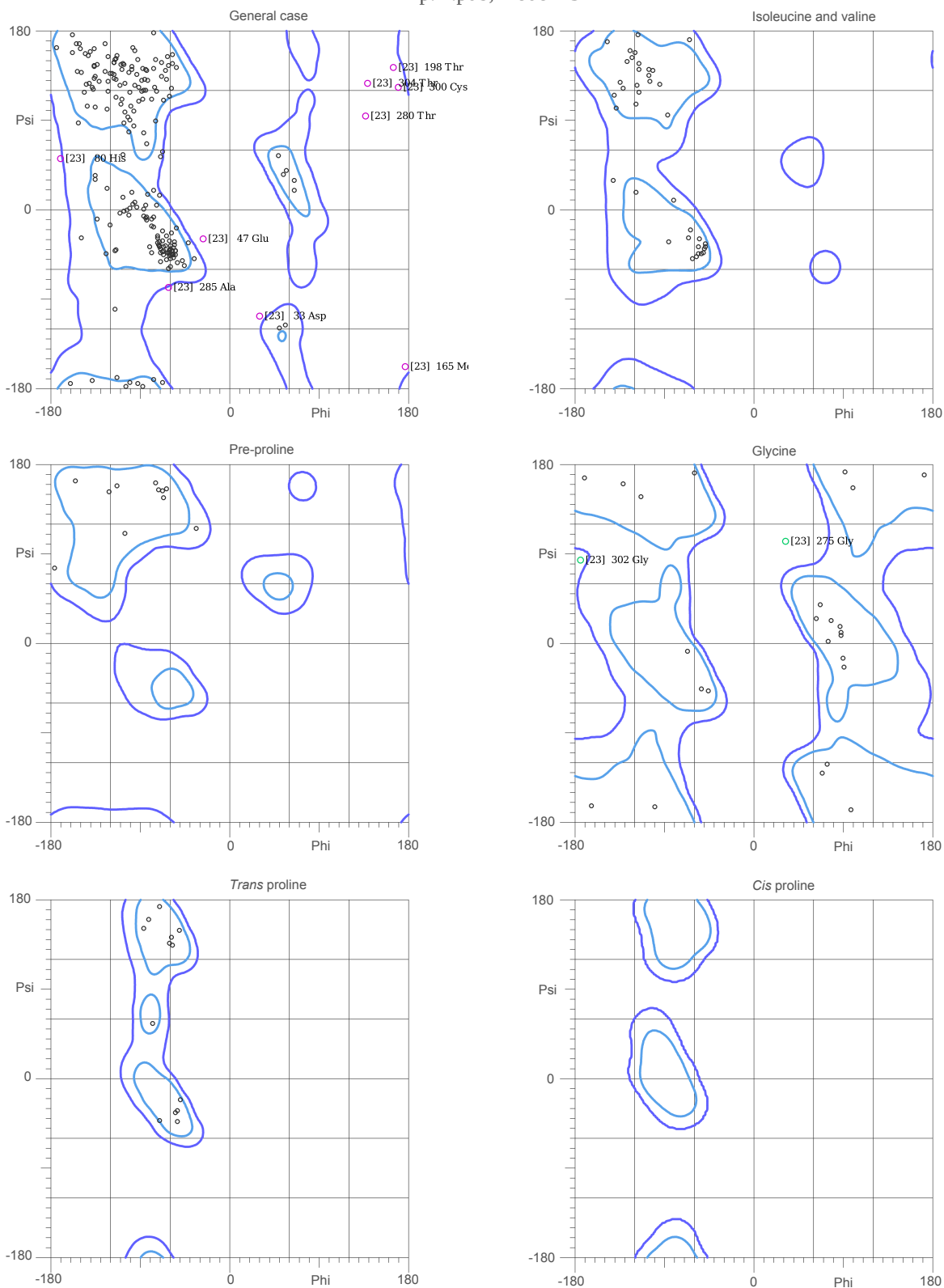
86.5% (263/304) of all residues were in favored (98%) regions.
96.1% (292/304) of all residues were in allowed (>99.8%) regions.

There were 12 outliers (phi, psi):

[22] 48 Asp (160.0, -38.1)	[22] 277 Asn (10.4, 72.2)
[22] 63 Asn (-17.8, -57.4)	[22] 299 Gln (-57.4, 1.7)
[22] 73 Val (-57.5, -166.8)	[22] 302 Gly (158.4, 38.4)
[22] 100 Lys (-50.1, 89.2)	[22] 304 Thr (131.9, 135.4)
[22] 143 Gly (17.9, 48.7)	
[22] 153 Asp (-57.9, 94.5)	
[22] 154 Tyr (28.6, -62.7)	
[22] 273 Gln (-71.8, -85.2)	

MolProbity Ramachandran analysis

7mhp.H.pdb, model 23



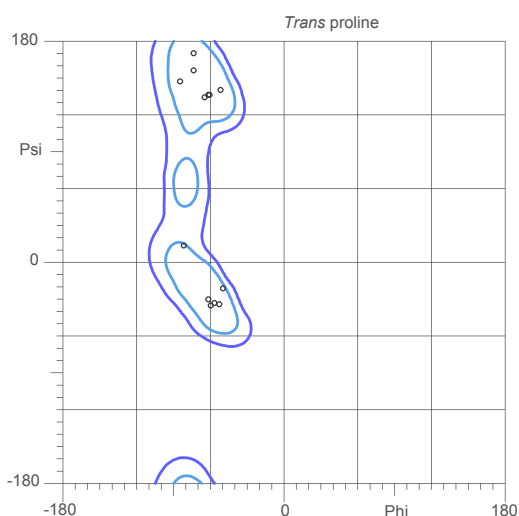
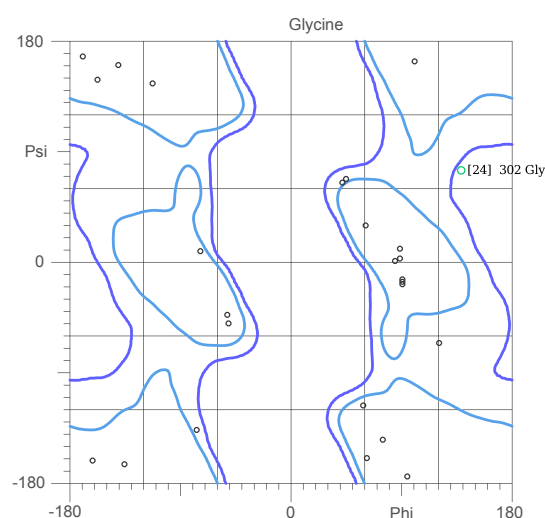
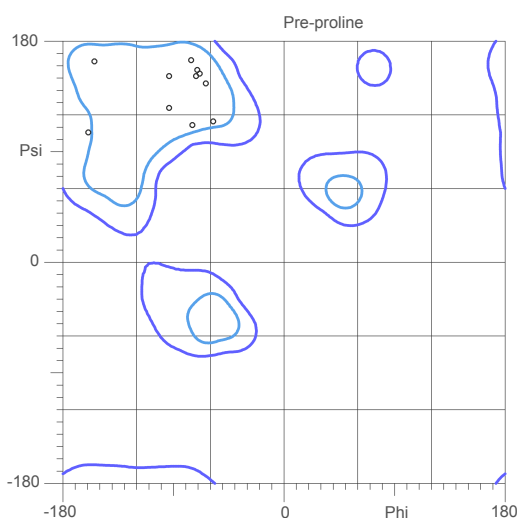
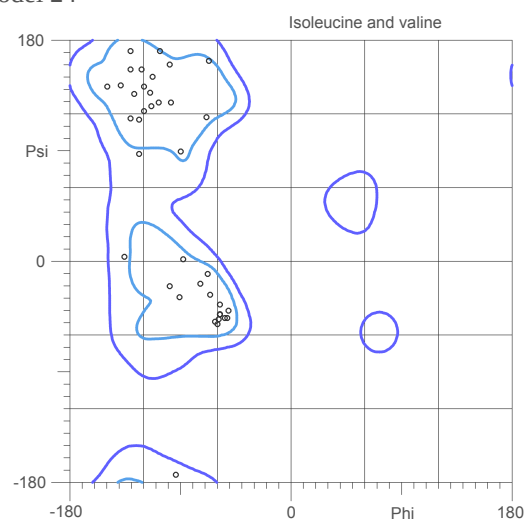
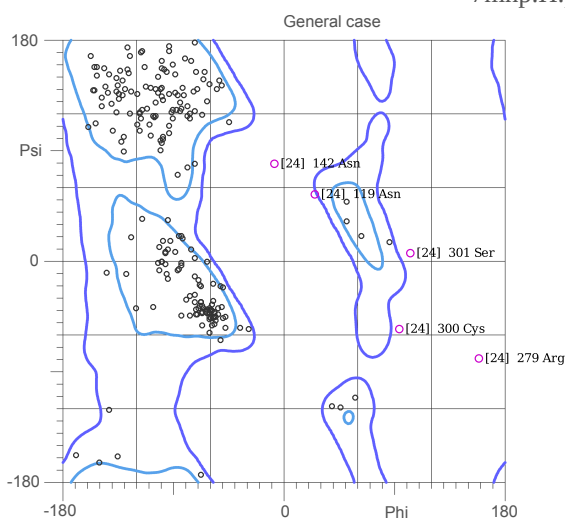
88.8% (270/304) of all residues were in favored (98%) regions.
96.4% (293/304) of all residues were in allowed (>99.8%) regions.

There were 11 outliers (phi, psi):

[23] 33 Asp (30.4, -107.6)	[23] 300 Cys (170.7, 124.5)
[23] 47 Glu (-27.9, -29.5)	[23] 302 Gly (-175.9, 84.5)
[23] 80 His (-171.5, 52.9)	[23] 304 Thr (139.7, 128.2)
[23] 165 Met (177.6, -158.8)	
[23] 198 Thr (165.0, 144.7)	
[23] 275 Gly (32.3, 103.7)	
[23] 280 Thr (137.8, 95.4)	
[23] 285 Ala (-62.8, -78.4)	

MolProbity Ramachandran analysis

7mhp.H.pdb, model 24



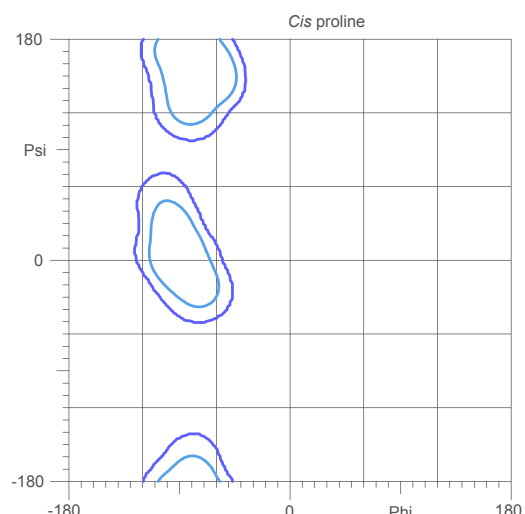
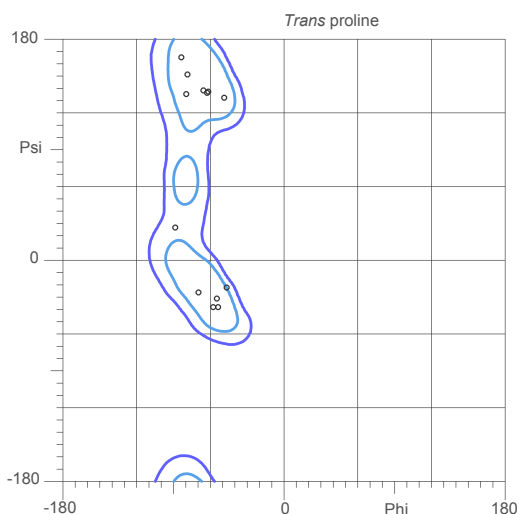
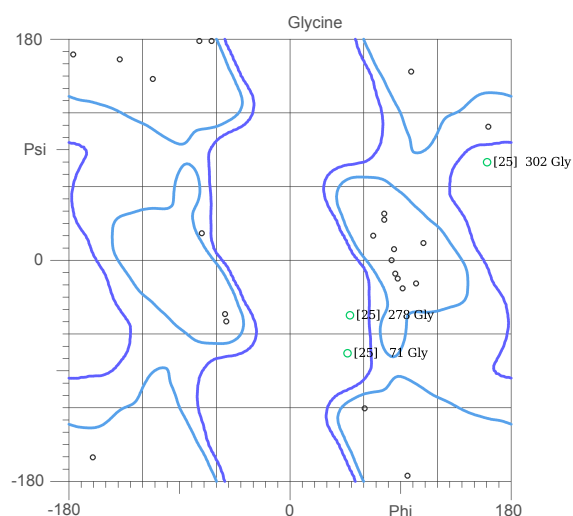
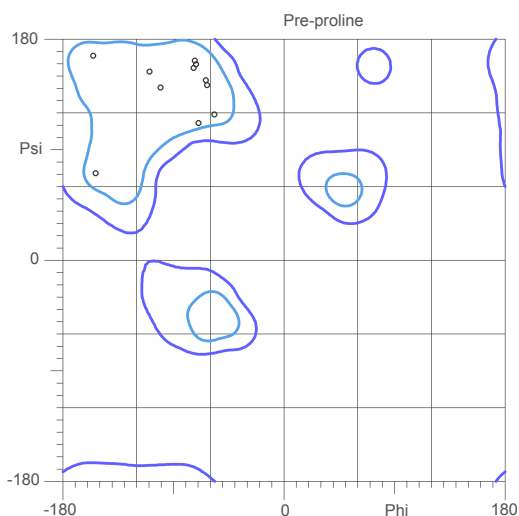
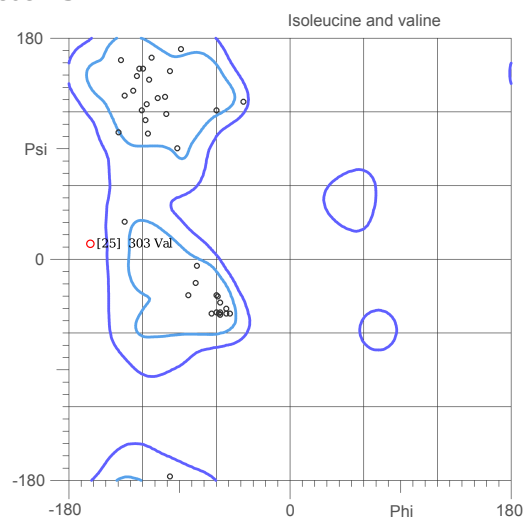
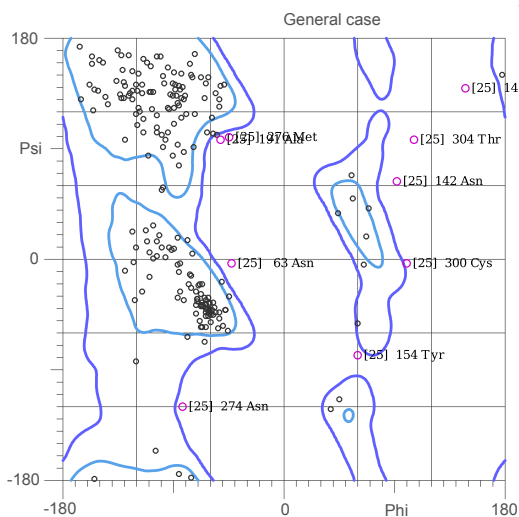
88.2% (268/304) of all residues were in favored (98%) regions.
98.0% (298/304) of all residues were in allowed (>99.8%) regions.

There were 6 outliers (phi, psi):

[24] 119 Asn (25.2, 55.9)
[24] 142 Asn (-8.5, 81.0)
[24] 279 Arg (159.6, -79.5)
[24] 300 Cys (95.0, -55.3)
[24] 301 Ser (103.1, 7.4)
[24] 302 Gly (139.6, 75.5)

MolProbity Ramachandran analysis

7mhp.H.pdb, model 25



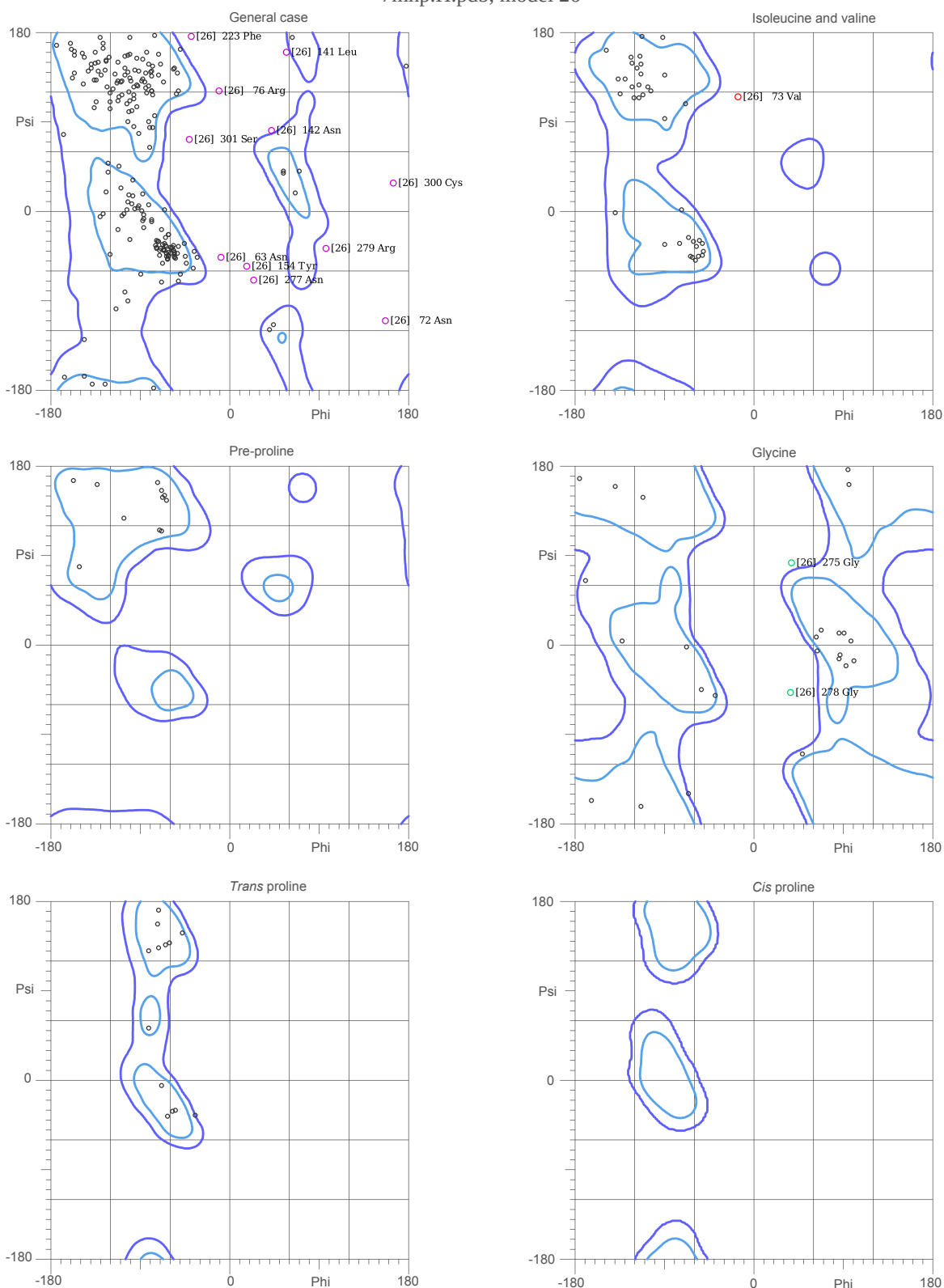
86.8% (264/304) of all residues were in favored (98%) regions.
95.7% (291/304) of all residues were in allowed (>99.8%) regions.

There were 13 outliers (phi, psi):

[25] 63 Asn (-43.7, -3.2)	[25] 278 Gly (49.8, -46.0)
[25] 71 Gly (47.5, -76.7)	[25] 300 Cys (100.1, -3.6)
[25] 141 Leu (148.2, 140.4)	[25] 302 Gly (161.6, 80.8)
[25] 142 Asn (92.5, 64.6)	[25] 303 Val (-163.3, 13.6)
[25] 154 Tyr (60.4, -78.6)	[25] 304 Thr (106.8, 98.3)
[25] 191 Ala (-52.4, 98.4)	
[25] 274 Asn (-83.1, -120.8)	
[25] 276 Met (-45.8, 100.8)	

MolProbity Ramachandran analysis

7mhp.H.pdb, model 26



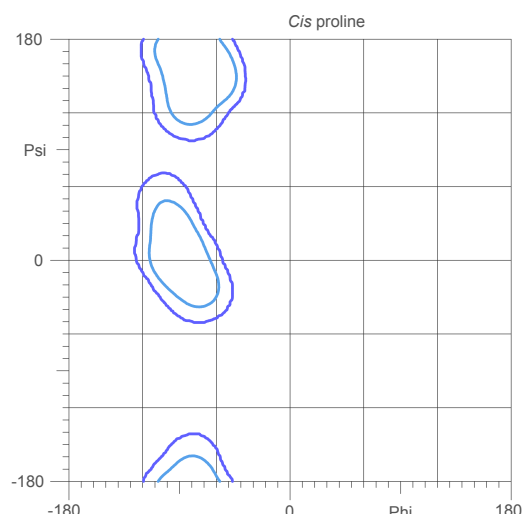
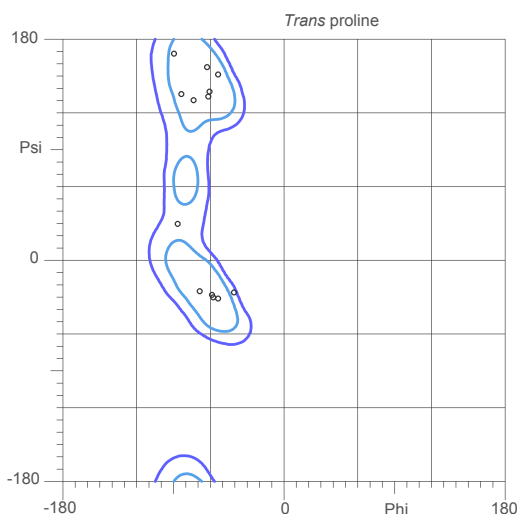
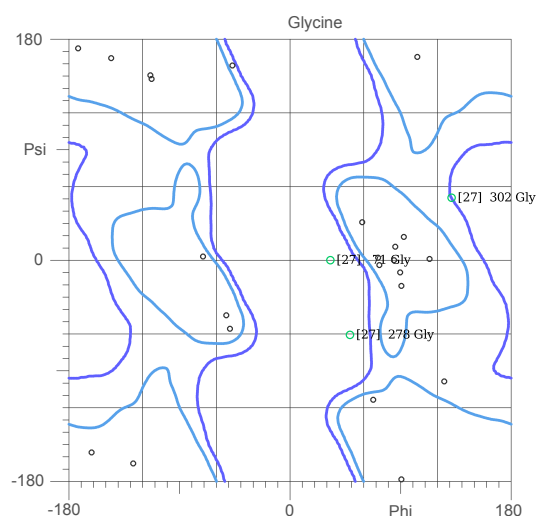
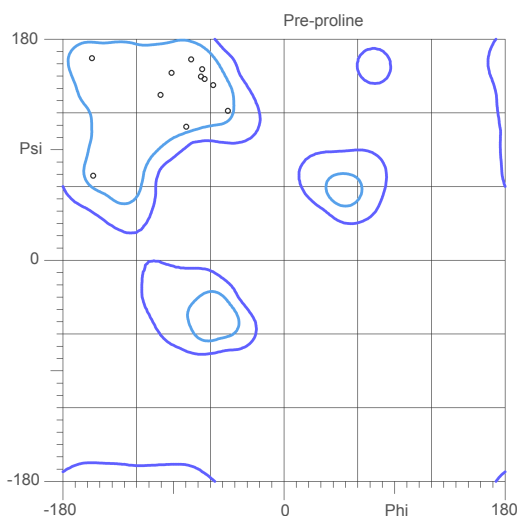
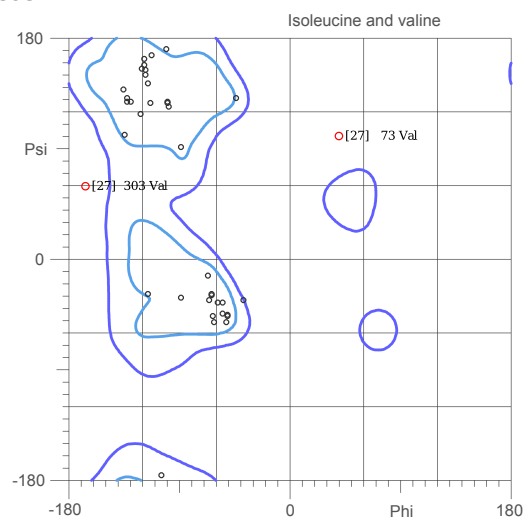
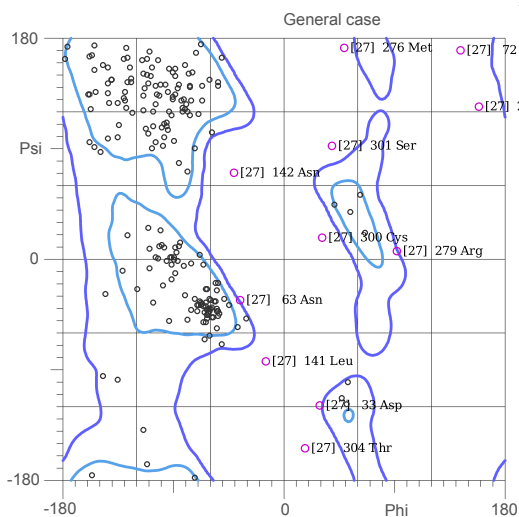
85.2% (259/304) of all residues were in favored (98%) regions.
95.4% (290/304) of all residues were in allowed (>99.8%) regions.

There were 14 outliers (phi, psi):

[26] 63 Asn (-9.5, -46.5)	[26] 275 Gly (38.3, 83.9)
[26] 72 Asn (157.7, -110.1)	[26] 277 Asn (24.4, -69.5)
[26] 73 Val (-16.8, 117.0)	[26] 278 Gly (37.6, -48.2)
[26] 76 Arg (-11.2, 122.6)	[26] 279 Arg (97.9, -37.0)
[26] 141 Leu (57.8, 162.0)	[26] 300 Cys (165.1, 29.2)
[26] 142 Asn (42.3, 82.5)	[26] 301 Ser (-41.1, 73.3)
[26] 154 Tyr (17.3, -56.0)	
[26] 223 Phe (-39.6, 177.6)	

MolProbity Ramachandran analysis

7mhp.H.pdb, model 27



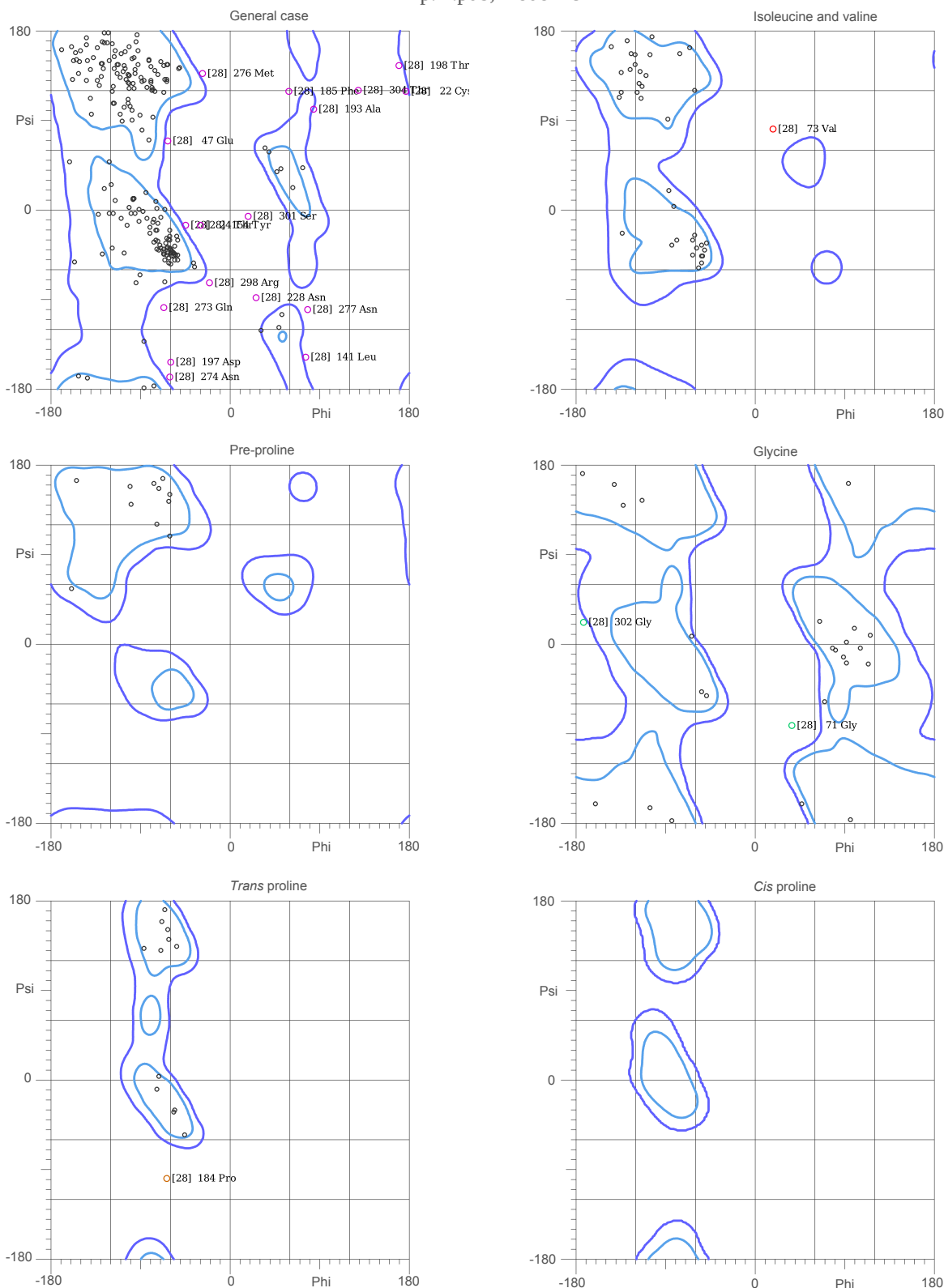
84.9% (258/304) of all residues were in favored (98%) regions.
94.7% (288/304) of all residues were in allowed (>99.8%) regions.

There were 16 outliers (phi, psi):

[27] 33 Asp (29.5, -119.1)	[27] 278 Gly (49.2, -61.7)
[27] 63 Asn (-36.1, -33.7)	[27] 279 Arg (92.2, 7.5)
[27] 71 Gly (33.2, -0.8)	[27] 280 Thr (159.5, 126.0)
[27] 72 Asn (144.8, 171.1)	[27] 300 Cys (31.7, 19.0)
[27] 73 Val (41.0, 101.3)	[27] 301 Ser (39.5, 93.1)
[27] 141 Leu (-15.4, -83.3)	[27] 302 Gly (132.5, 51.5)
[27] 142 Asn (-41.5, 72.0)	[27] 303 Val (-167.5, 60.6)
[27] 276 Met (49.9, 173.6)	[27] 304 Thr (17.3, -154.1)

MolProbity Ramachandran analysis

7mhp.H.pdb, model 28



85.2% (259/304) of all residues were in favored (98%) regions.
93.1% (283/304) of all residues were in allowed (>99.8%) regions.

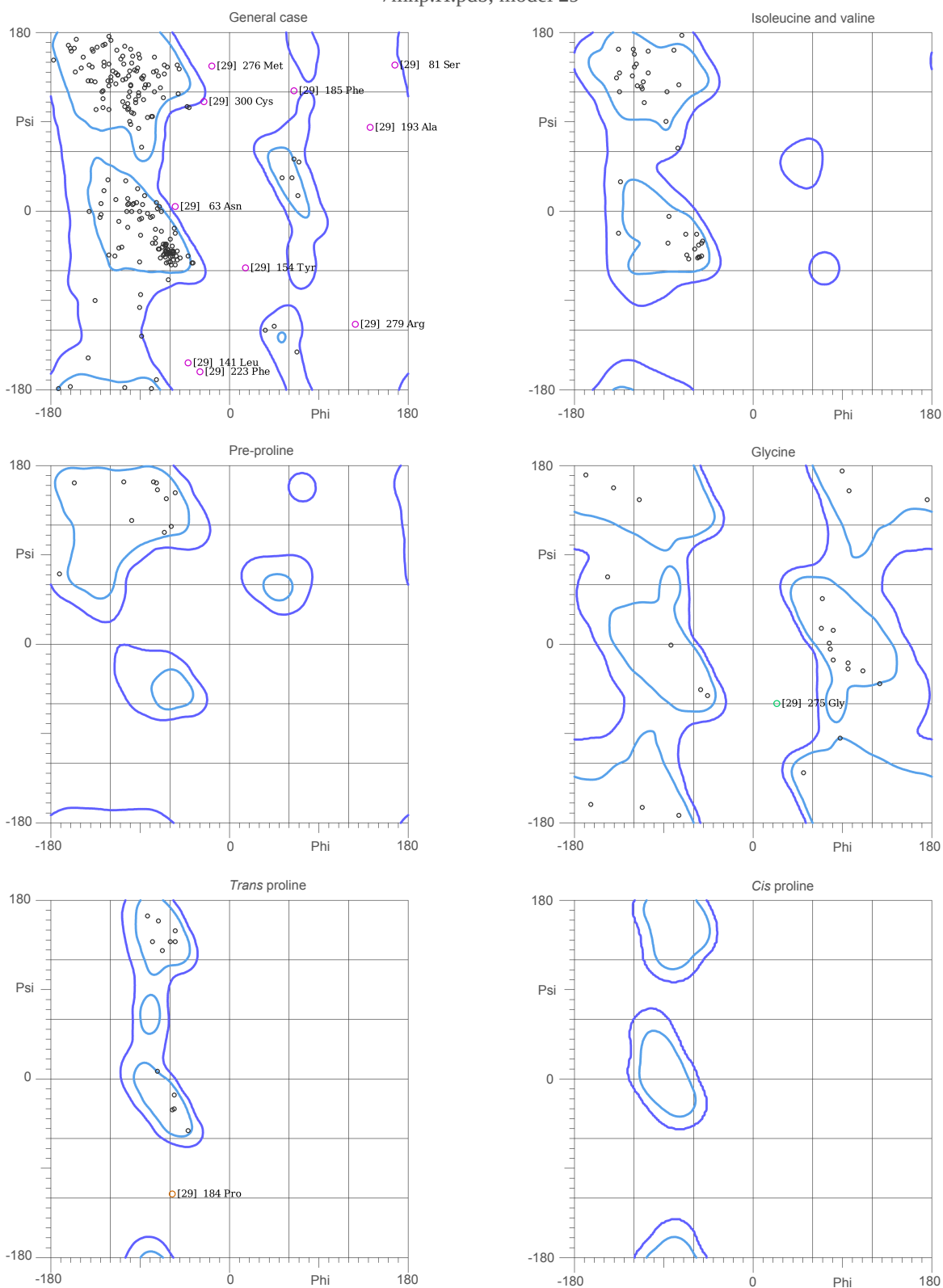
There were 21 outliers (phi, psi):

[28] 22 Cys (177.4, 120.2)	[28] 185 Phe (59.7, 120.4)
[28] 24 Thr (-45.5, -15.5)	[28] 193 Ala (84.3, 102.9)
[28] 47 Glu (-63.4, 70.1)	[28] 197 Asp (-60.4, -153.7)
[28] 71 Gly (37.8, -82.2)	[28] 198 Thr (170.3, 146.7)
[28] 73 Val (18.4, 82.2)	[28] 228 Asn (26.1, -88.0)
[28] 141 Leu (76.2, -148.7)	[28] 273 Gln (-67.4, -98.6)
[28] 154 Tyr (-30.2, -15.5)	[28] 274 Asn (-61.6, -168.2)
[28] 184 Pro (-64.4, -99.1)	[28] 276 Met (-28.5, 139.0)

[28] 277 Asn (78.6, -100.7)
[28] 298 Arg (-21.6, -73.9)
[28] 301 Ser (18.9, -6.7)
[28] 302 Gly (-173.7, 22.3)
[28] 304 Thr (129.7, 121.4)

MolProbity Ramachandran analysis

7mhp.H.pdb, model 29



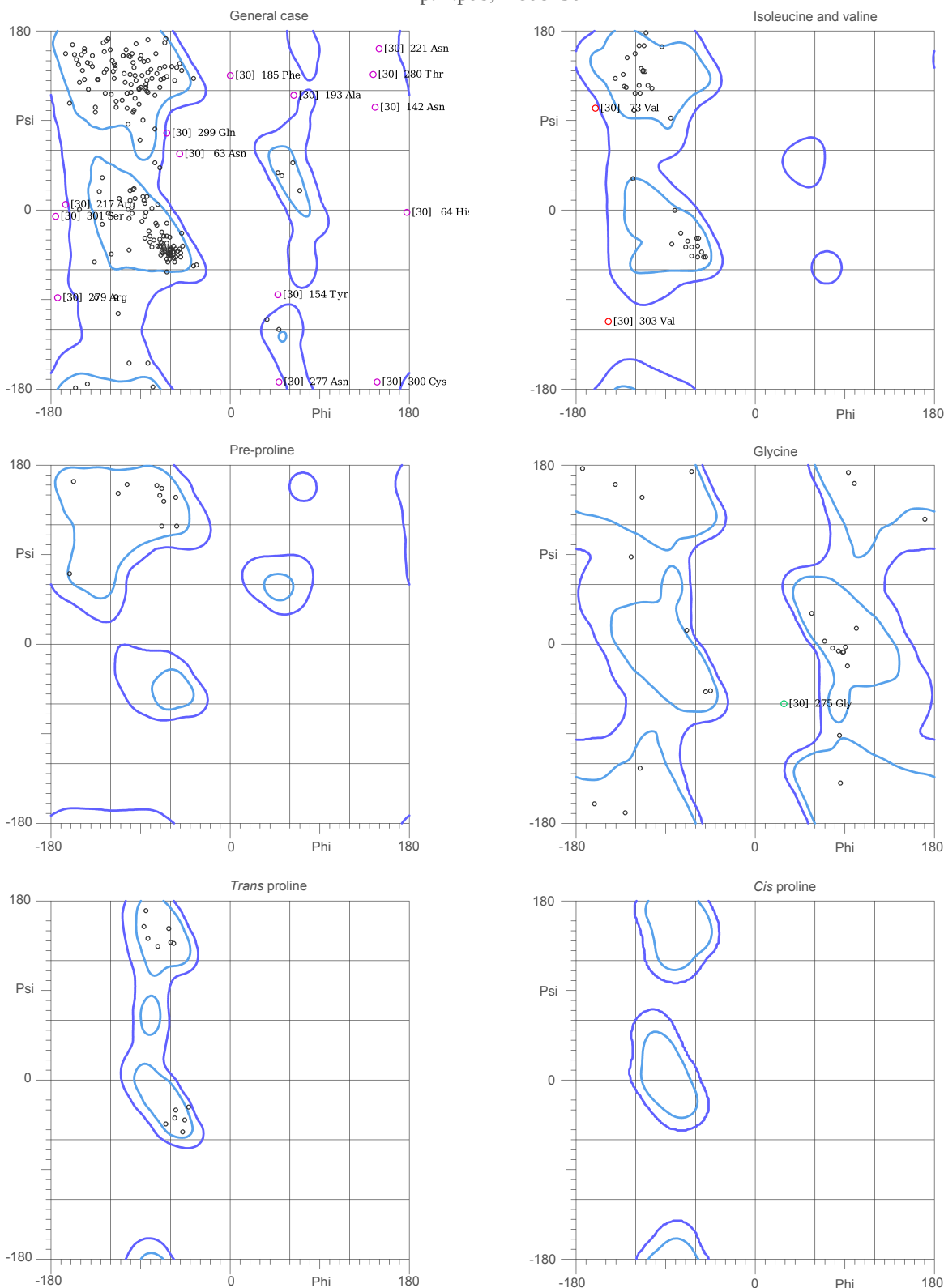
87.2% (265/304) of all residues were in favored (98%) regions.
96.1% (292/304) of all residues were in allowed (>99.8%) regions.

There were 12 outliers (phi, psi):

[29] 63 Asn (-55.7, 5.2)	[29] 275 Gly (24.6, -60.3)
[29] 81 Ser (167.8, 148.1)	[29] 276 Met (-18.2, 147.4)
[29] 141 Leu (-42.7, -153.9)	[29] 279 Arg (127.6, -114.1)
[29] 154 Tyr (16.9, -58.0)	[29] 300 Cys (-26.8, 111.3)
[29] 184 Pro (-58.3, -116.3)	
[29] 185 Phe (65.8, 122.6)	
[29] 193 Ala (142.5, 86.0)	
[29] 223 Phe (-30.7, -162.1)	

MolProbity Ramachandran analysis

7mhp.H.pdb, model 30



86.2% (262/304) of all residues were in favored (98%) regions.
94.4% (287/304) of all residues were in allowed (>99.8%) regions.

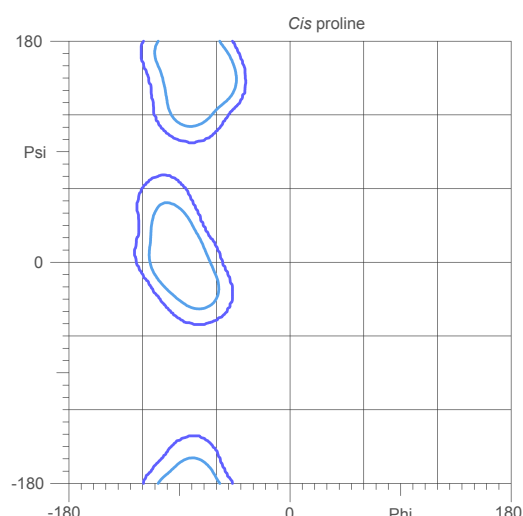
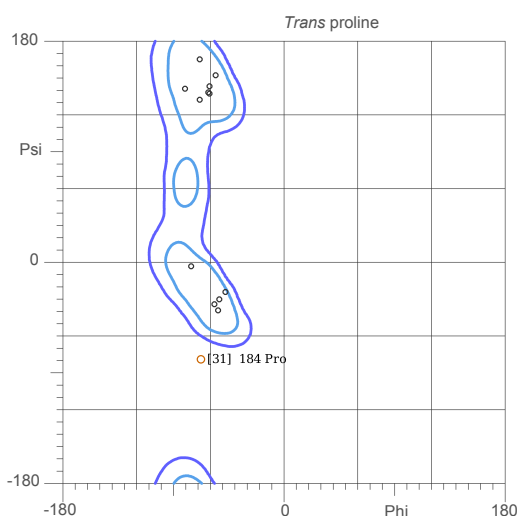
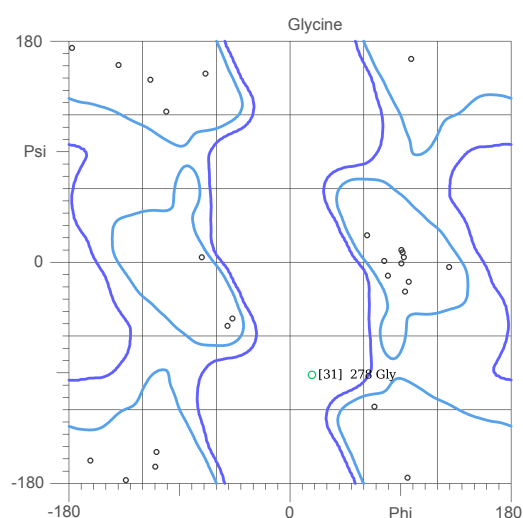
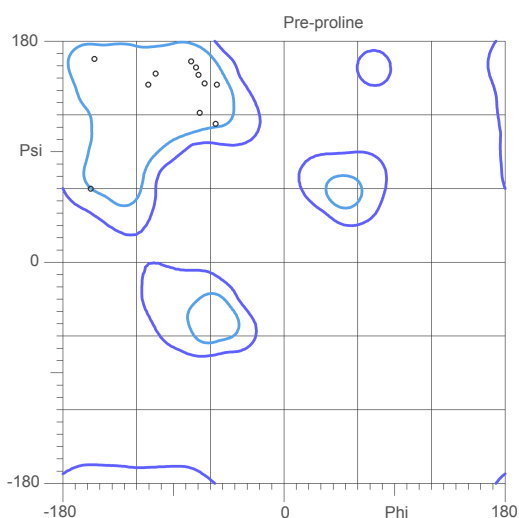
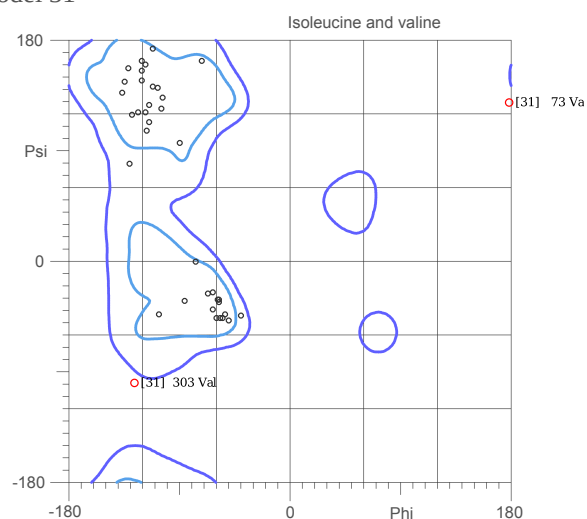
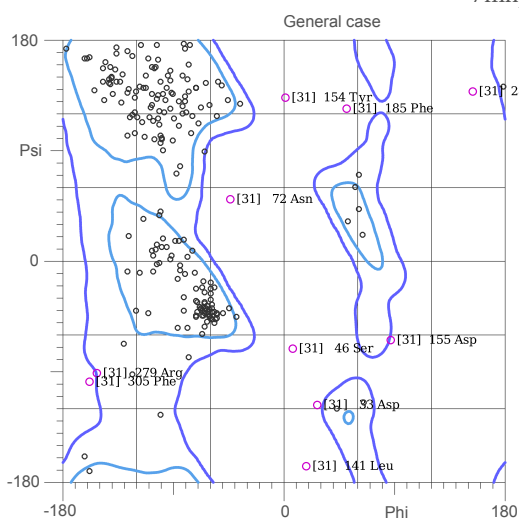
[30] 303 Val (-148.8, -112.9)

There were 17 outliers (phi, psi):

[30] 63 Asn (-51.2, 57.4)	[30] 221 Asn (150.5, 163.8)
[30] 64 His (178.2, -2.2)	[30] 275 Gly (29.8, -60.6)
[30] 73 Val (-161.9, 103.6)	[30] 277 Asn (49.4, -173.8)
[30] 142 Asn (146.5, 104.7)	[30] 279 Arg (-174.9, -88.3)
[30] 154 Tyr (49.0, -85.7)	[30] 280 Thr (144.7, 137.6)
[30] 185 Phe (-0.8, 136.3)	[30] 299 Gln (-64.8, 78.3)
[30] 193 Ala (64.8, 116.6)	[30] 300 Cys (148.6, -173.1)
[30] 217 Arg (-166.0, 6.4)	[30] 301 Ser (-176.9, -6.4)

MolProbity Ramachandran analysis

7mhp.H.pdb, model 31



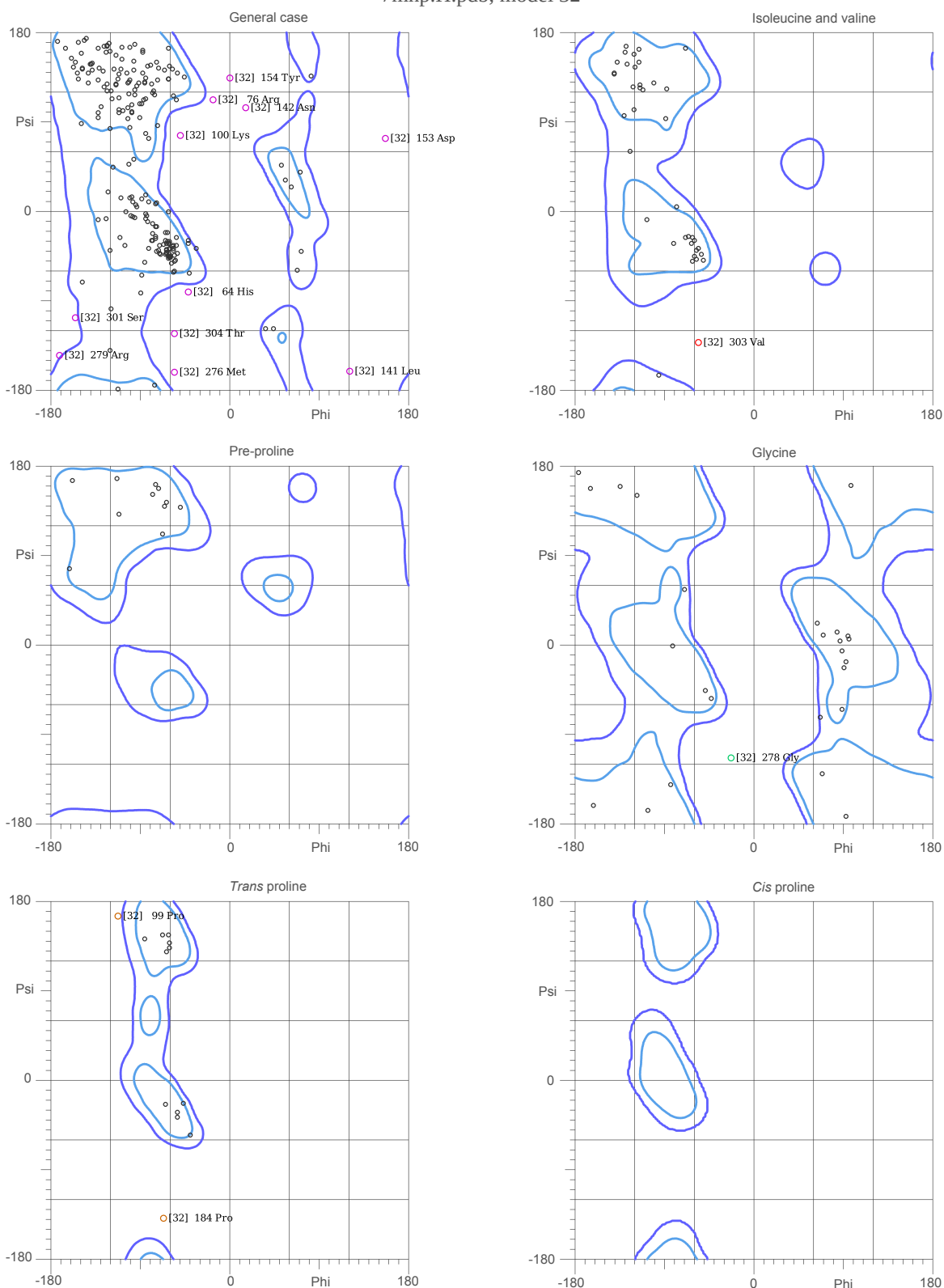
89.1% (271/304) of all residues were in favored (98%) regions.
95.4% (290/304) of all residues were in allowed (>99.8%) regions.

There were 14 outliers (phi, psi):

[31] 33 Asp (27.8, -117.5)	[31] 185 Phe (51.4, 125.7)
[31] 46 Ser (7.9, -71.5)	[31] 278 Gly (18.6, -92.5)
[31] 72 Asn (-44.7, 51.1)	[31] 279 Arg (-153.3, -91.4)
[31] 73 Val (179.8, 130.7)	[31] 280 Thr (154.4, 139.3)
[31] 141 Leu (18.4, -167.6)	[31] 303 Val (-127.3, -99.3)
[31] 154 Tyr (1.4, 134.3)	[31] 305 Phe (-159.7, -98.3)
[31] 155 Asp (87.5, -64.4)	
[31] 184 Pro (-68.0, -79.5)	

MolProbity Ramachandran analysis

7mhp.H.pdb, model 32



85.5% (260/304) of all residues were in favored (98%) regions.

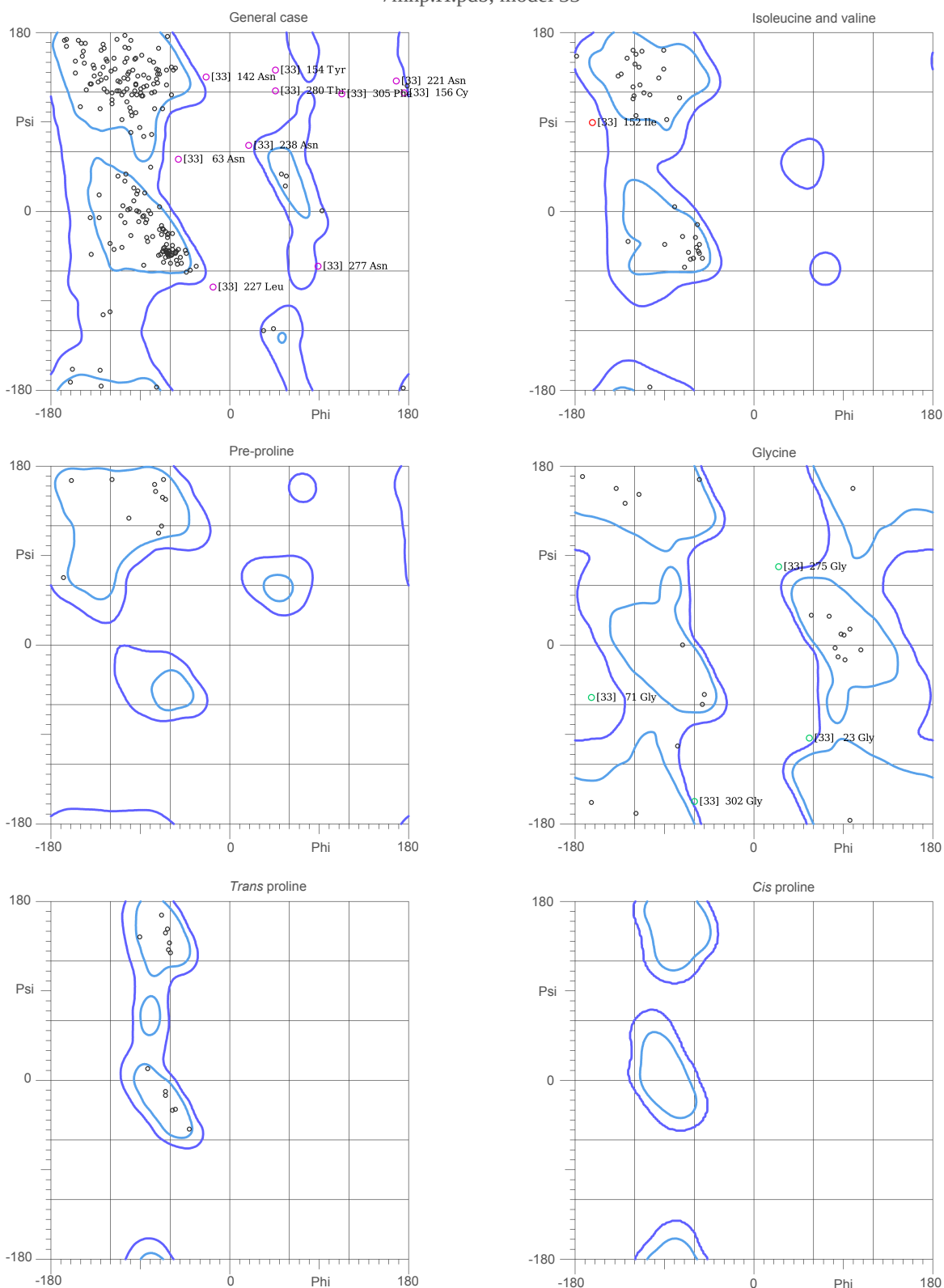
95.1% (289/304) of all residues were in allowed (>99.8%) regions.

There were 15 outliers (phi, psi):

[32] 64 His (-42.6, -81.5)	[32] 184 Pro (-67.1, -139.1)
[32] 76 Arg (-17.4, 113.1)	[32] 276 Met (-56.6, -162.9)
[32] 99 Pro (-113.4, 166.7)	[32] 278 Gly (-23.9, -114.1)
[32] 100 Lys (-50.9, 77.1)	[32] 279 Arg (-172.4, -145.1)
[32] 141 Leu (121.6, -162.0)	[32] 301 Ser (-156.0, -107.5)
[32] 142 Asn (16.1, 105.6)	[32] 303 Val (-56.1, -132.9)
[32] 153 Asp (158.0, 74.4)	[32] 304 Thr (-56.3, -123.1)
[32] 154 Tyr (-0.1, 135.8)	

MolProbity Ramachandran analysis

7mhp.H.pdb, model 33



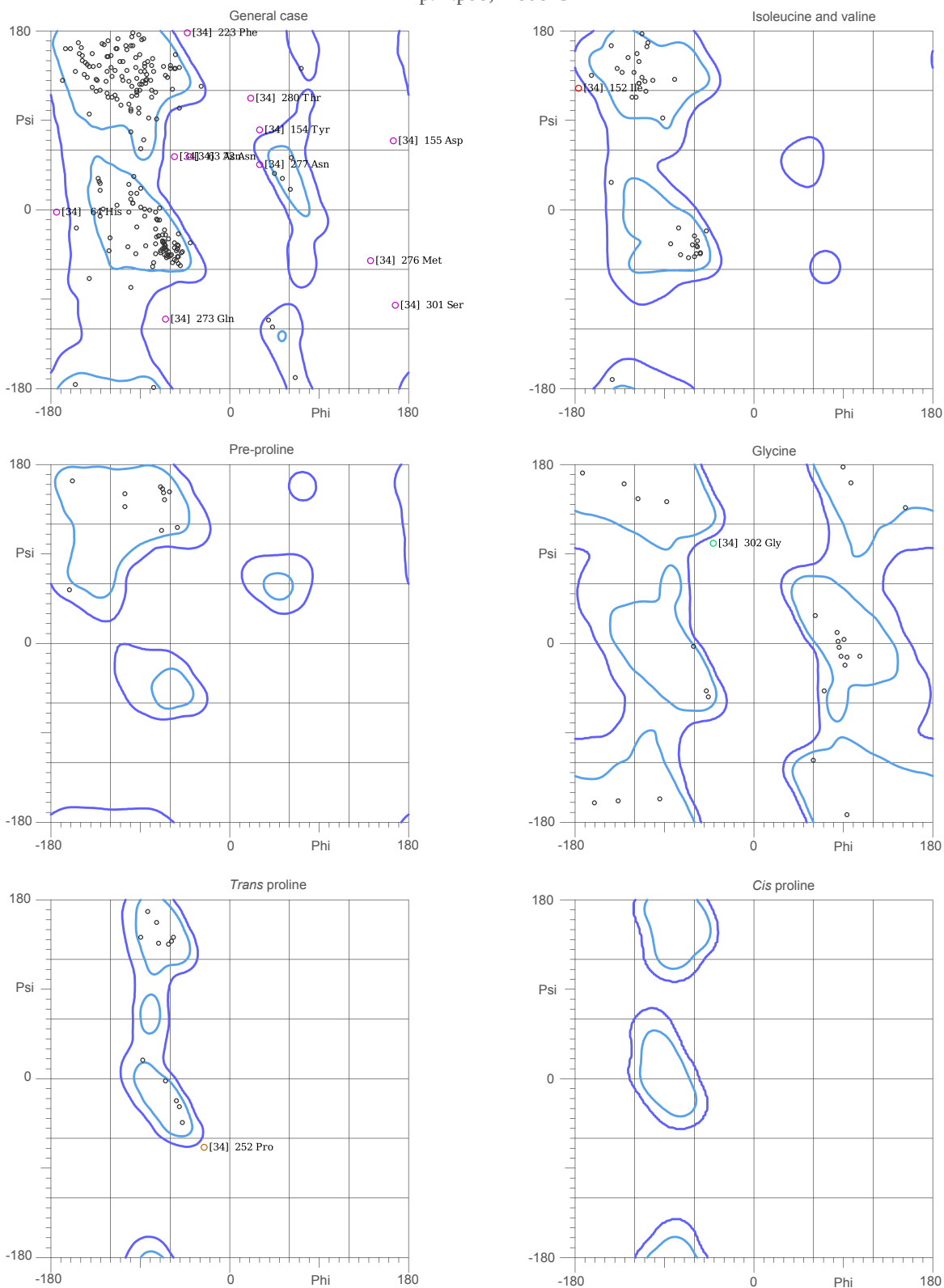
87.5% (266/304) of all residues were in favored (98%) regions.
95.1% (289/304) of all residues were in allowed (>99.8%) regions.

There were 15 outliers (phi, psi):

[33] 23 Gly (57.0, -95.0)	[33] 227 Leu (-17.6, -76.0)
[33] 63 Asn (-52.7, 53.9)	[33] 238 Asn (19.3, 67.5)
[33] 71 Gly (-164.0, -53.2)	[33] 275 Gly (25.9, 79.8)
[33] 142 Asn (-24.1, 136.3)	[33] 277 Asn (89.5, -55.1)
[33] 152 Ile (-163.9, 90.3)	[33] 280 Thr (46.6, 122.9)
[33] 154 Tyr (46.3, 143.1)	[33] 302 Gly (-60.4, -159.0)
[33] 156 Cys (176.6, 120.3)	[33] 305 Phe (113.5, 119.4)
[33] 221 Asn (168.2, 132.4)	

MolProbity Ramachandran analysis

7mhp.H.pdb, model 34



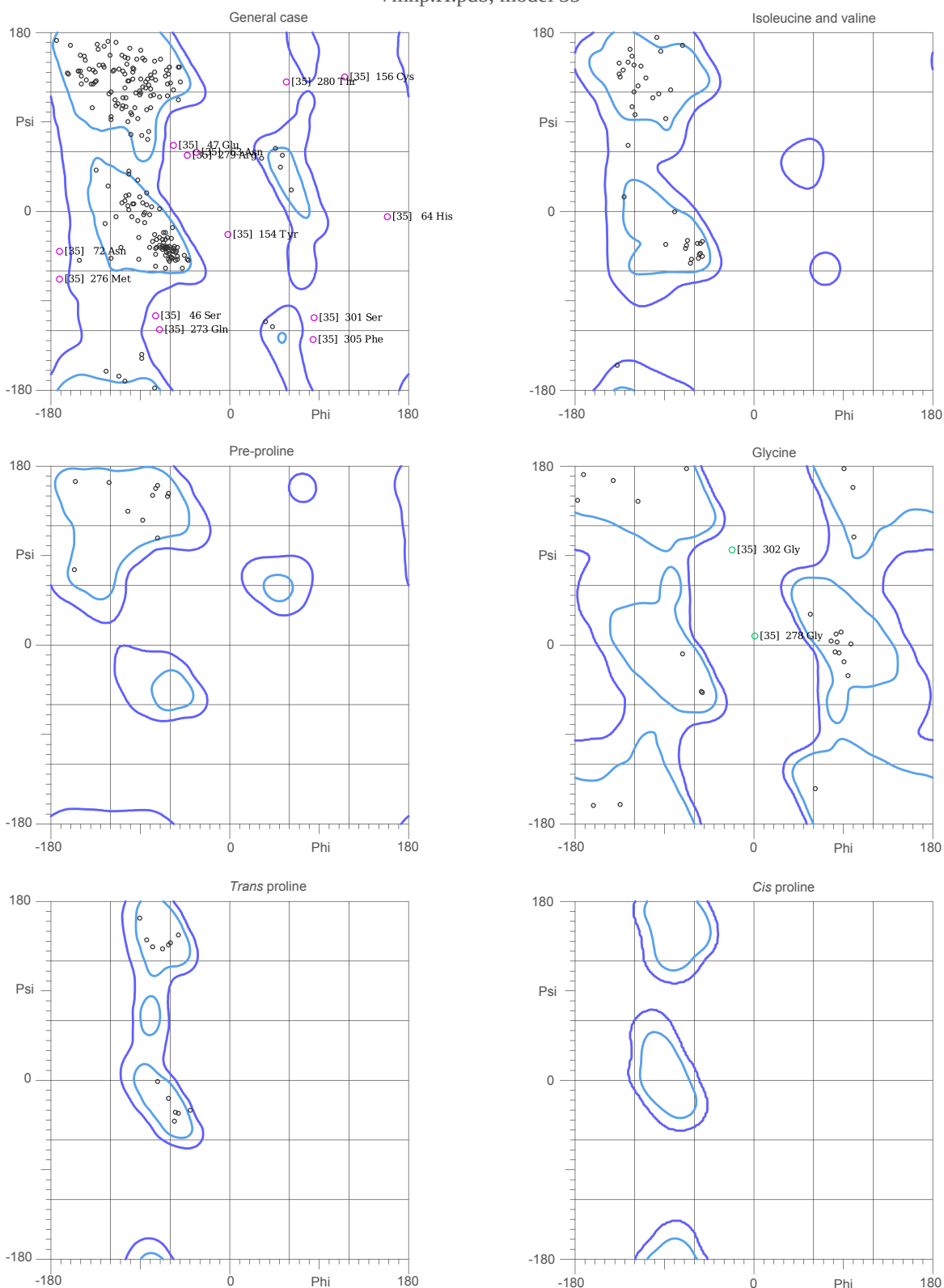
88.8% (270/304) of all residues were in favored (98%) regions.
95.4% (290/304) of all residues were in allowed (>99.8%) regions.

There were 14 outliers (phi, psi):

[34] 63 Asn (-56.4, 54.9)	[34] 273 Gln (-65.5, -110.6)
[34] 64 His (-175.7, -2.7)	[34] 276 Met (142.4, -51.9)
[34] 72 Asn (-41.2, 54.1)	[34] 277 Asn (30.8, 46.5)
[34] 152 Ile (-177.6, 123.7)	[34] 280 Thr (21.8, 113.8)
[34] 154 Tyr (30.5, 81.5)	[34] 301 Ser (167.8, -96.5)
[34] 155 Asp (165.1, 70.5)	[34] 302 Gly (-41.7, 101.3)
[34] 223 Phe (-44.0, 179.6)	
[34] 252 Pro (-26.4, -69.5)	

MolProbity Ramachandran analysis

7mhp.H.pdb, model 35



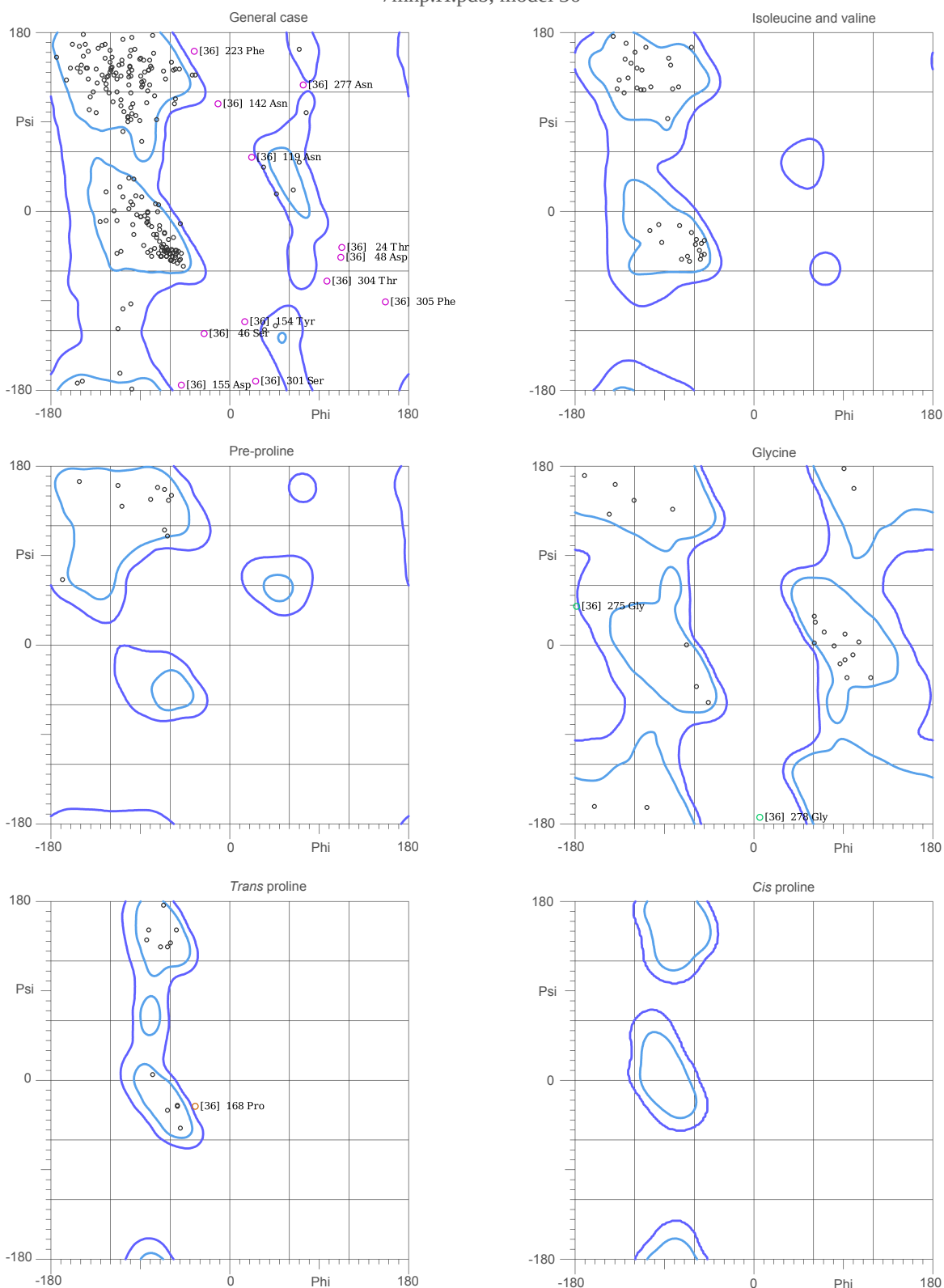
88.5% (269/304) of all residues were in favored (98%) regions.
95.1% (289/304) of all residues were in allowed (>99.8%) regions.

There were 15 outliers (phi, psi):

[35] 46 Ser (-75.7, -105.2)	[35] 276 Met (-172.9, -68.9)
[35] 47 Glu (-57.7, 67.9)	[35] 278 Gly (1.4, 10.0)
[35] 63 Asn (-34.3, 60.8)	[35] 279 Arg (-43.1, 57.9)
[35] 64 His (159.7, -5.6)	[35] 280 Thr (58.0, 131.6)
[35] 72 Asn (-172.3, -40.8)	[35] 301 Ser (86.0, -107.9)
[35] 154 Tyr (-2.8, -23.8)	[35] 302 Gly (-22.9, 96.8)
[35] 156 Cys (116.6, 136.0)	[35] 305 Phe (84.6, -129.1)
[35] 273 Gln (-71.2, -119.5)	

MolProbity Ramachandran analysis

7mhp.H.pdb, model 36



88.5% (269/304) of all residues were in favored (98%) regions.
95.1% (289/304) of all residues were in allowed (>99.8%) regions.

There were 15 outliers (phi, psi):

[36] 24 Thr (113.2, -36.6)	[36] 223 Phe (-36.3, 162.3)
[36] 46 Ser (-26.0, -123.5)	[36] 275 Gly (-179.4, 39.8)
[36] 48 Asp (112.1, -46.8)	[36] 277 Asn (74.8, 128.9)
[36] 119 Asn (22.2, 55.1)	[36] 278 Gly (6.5, -174.1)
[36] 142 Asn (-12.8, 109.2)	[36] 301 Ser (26.9, -171.0)
[36] 154 Tyr (15.7, -111.4)	[36] 304 Thr (98.7, -70.1)
[36] 155 Asp (-49.4, -175.9)	[36] 305 Phe (157.8, -91.2)
[36] 168 Pro (-35.4, -26.4)	